Heterogeneity-Aware Self-Supervised Representation Learning and Stability-Aware Clustering for Unlabeled Scanned Document Archives

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***Abstract*—Unlabeled scanned archives contain heterogeneous evidence: page geometry, visual content, and OCR-derived text may be informative for some documents and absent or unreliable for others. This paper presents Heterogeneity-Aware Multi-View Contrastive Clustering (HAMV-CC), a label-free representation and clustering framework that uses availability masks, reliability-weighted multi-view fusion, instance contrast, cross-view agreement, cluster consistency, and stability-aware cluster-count selection. The study deliberately separates implemented behavior from empirical evidence. The baseline system provides a sixteen-dimensional interpretable descriptor, optional ResNet50 features, a feature-space self-supervised encoder, conventional clustering, and an auditable desktop workflow. The new HAMV-CC implementation was executed for ten matched seeds on a controlled 180-sample, three-view synthetic fixture with approximately 12% independently missing view entries. At $k = 3$, HAMV-CC obtained ARI 0.8882 ± 0.0960, NMI 0.8769 ± 0.1006, and pairwise seed ARI stability 0.8281 ± 0.1037. A masked reliability-weighted concatenation control obtained ARI/NMI $= 1.0000/1.0000$ on the same fixture; this negative comparison is retained explicitly and is not interpreted as evidence of superiority. In a separate 100-image external archive, ten-seed execution of the implemented baseline showed lower internal geometry and assignment stability for feature-space SSL than for the handcrafted representation. These results validate the reproducibility and falsifiability of the pipeline, but do not establish HAMV-CC superiority, semantic accuracy, or cross-domain generalization. The manuscript therefore provides a frozen ablation, missing-view, independent-audit, external-benchmark, and efficiency protocol as the required final validation gate.**

Evidence status: the supplied private case study supports implementation traceability and single-run clustering evidence. Repeated-seed comparisons, independent human labels, and external benchmark results are not claimed unless explicitly populated in the final experimental record.

***Index Terms*—self-supervised learning · contrastive clustering · document image analysis · heterogeneous archives · multi-view representation learning · cluster stability · OCR · unsupervised learning**

I. Introduction

Digitization efforts routinely convert physical print archives — historical advertisements, magazines, packaging, forms, and correspondence — into large collections of scanned raster images. Unlike born-digital documents, these scans arrive with no reliable metadata: filenames are often arbitrary, folder structure reflects the order of scanning rather than content, and no ground-truth labels exist for genre, document type, or subject matter. Manually sorting even a few thousand such images by content is prohibitively time-consuming, yet many downstream tasks — archival cataloguing, provenance research, or dataset curation for further machine learning — require exactly this kind of organization. Supervised classification is unattractive in this setting because it presumes a labelled training set that, by definition, does not exist for a fresh, unlabeled archive.

This motivates an unsupervised approach: learn a representation space from measurable document and visual features, partition that space into clusters, and then attach human-readable labels using lightweight, interpretable signals. The supplied application follows this strategy by combining handcrafted descriptors, optional ResNet50 features, and a SimCLR-style feature-space MLP.

The resulting embeddings are scaled, optionally reduced with PCA, and clustered with one of eight classical algorithms. The candidate cluster count is selected by a simple internal-metric voting rule. Two complementary labelling stages then enrich the clusters: a handcrafted document-type analyzer and an optional OCR/ResNet50 genre-and-brand classifier. The entire workflow is exposed through a PyQt5 desktop application with progress reporting, cancellation, visualization, and export.

The technical scope of this work is deliberately separated into implemented capabilities and evaluation requirements. The implemented system provides: (1) a compact document representation combining handcrafted and optional pretrained visual features; (2) feature-space self-supervised learning with a lightweight MLP and NT-Xent loss; (3) a configurable registry of eight clustering procedures, including bootstrap consensus clustering; (4) interpretable document-type analysis and optional OCR/ResNet50 semantic enrichment; (5) a PyQt5 desktop workflow with threaded execution, progress reporting, cancellation, visualization, session persistence, and export; and (6) a reproducibility and audit specification that makes the evidence boundary explicit.

The central systems question is how to connect document feature extraction, self-supervised representation learning, clustering, semantic description, provenance, and human review without allowing heuristic labels to masquerade as ground truth. The study therefore separates three evidence layers: geometric cluster validity, assignment stability, and external human assessment. The principal evaluation questions are RQ1, whether self-supervised transformation changes cluster structure relative to fixed-feature baselines; RQ2, whether assignments remain stable under repeated seeds, resampling, and controlled

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scan degradation; and RQ3, whether clusters and post-hoc semantic descriptions are useful to human reviewers on an independently held-out audit subset. These are evaluation targets rather than completed findings. This separation is important because internal validity measures describe geometry, semantic heuristics provide auxiliary labels, and human review supplies external evidence; none is sufficient on its own to establish semantic correctness.

II. Research Gap, Problem Formulation, and Contributions

Recent document representation methods demonstrate that self-supervised learning can remove part of the annotation burden in document understanding, while document clustering research increasingly explores multi-view consistency and clustering-aware objectives. However, archival scans differ from curated document benchmarks in three important ways. First, the available evidence is heterogeneous: a page may be dominated by layout and geometry, visual imagery, or text, and OCR may fail selectively. Second, archive organization is an open-set discovery problem in which the number of meaningful groups is unknown. Third, a geometrically coherent cluster is not necessarily semantically meaningful. A method can therefore obtain a favorable silhouette score while grouping documents according to scanner artifacts, color balance, or acquisition conditions rather than archival semantics.

The supplied implementation provides a reproducible baseline but exposes precisely these gaps. Its active self-supervised path operates on a concatenated feature vector and uses independent Gaussian perturbations; the available image augmentation helper is not part of the executed training path. Its cluster-count selector is a heuristic weighted vote over internal indices, and its semantic labels are explicitly auxiliary rather than independent ground truth. These implementation facts are retained rather than obscured.

We formulate the archive as an unlabeled set $\mathcal{D} = \{d_i\}_{i=1}^{N}$, where each document may provide several partially observed views $x_i^{(m)}$, $m \in \{s, v, t\}$ for structural/handcrafted, visual, and textual evidence. The objective is to learn an embedding $h_i = f_\theta(\{x_i^{(m)}\})$ such that (i) semantically equivalent perturbations remain close, (ii) unreliable views do not dominate the representation, (iii) samples assigned to the same latent group exhibit assignment consistency, and (iv) the selected partition remains stable under resampling and realistic scan degradation.

The paper makes five contributions. First, it provides an implementation-traceable feature-space self-supervised baseline for heterogeneous scanned archives, including exact feature definitions, training behavior, failure handling, and export contracts. Second, it formulates HAMV-CC, a reliability-weighted multi-view extension that explicitly models missing or unreliable document evidence rather than treating every feature source as equally informative. Third, it couples instance-level contrastive learning with a cluster-consistency objective so that the representation is optimized for both invariance and partition structure. Fourth, it introduces stability-aware cluster-count selection that combines internal validity with repeated-assignment consistency and cluster-balance diagnostics. Fifth, it establishes an evidence hierarchy separating internal geometry, stochastic stability, and independently audited semantic validity, together with a reproducible experimental protocol that prevents pseudo-label leakage.

The first contribution is implemented in the supplied application. Contributions two through four constitute the journal-level methodological extension and must be experimentally validated before they are presented as demonstrated superiority. This distinction is essential: the manuscript does not convert a proposed algorithm into a claimed result merely by describing it.

A. Research Hypotheses

The enhanced study is organized around falsifiable hypotheses rather than a single aggregate score:

- H1: reliability-weighted multi-view fusion produces more semantically coherent embeddings than fixed concatenation under the same clustering procedure;
- H2: document-aware perturbations and cross-view consistency improve assignment stability under controlled scan degradation;
- H3: cluster-consistency regularization improves the agreement between internal geometry and independently audited semantic groups;
- H4: stability-aware cluster-count selection reduces partition variability relative to any single internal validity index; and
- H5: the proposed representation achieves these gains without an impractical increase in training time or peak memory.

These hypotheses define the minimum evidence required for a strong journal-level claim; failure to support one of them will be reported as a negative or null result rather than hidden through selective metric reporting.

III. Related Work

Contrastive self-supervised learning has become a standard route to label-free visual representations. SimCLR [1] showed that a simple framework — two stochastically augmented views of the same image pulled together in embedding space via a normalized temperature-scaled cross-entropy (NT-Xent) loss, with negatives drawn from the rest of the batch — can match supervised pretraining on downstream tasks when paired with a sufficiently expressive backbone and strong augmentation.

Related contrastive and non-contrastive alternatives include momentum-based negative sampling with a slowly updated key encoder [11] and Siamese predictor architectures that learn stable representations without explicit negative pairs [12]. The present implementation adopts SimCLR's in-batch-negative objective, but applies it to precomputed document features rather than directly to image pixels. This distinction makes the method substantially lighter than a conventional image-encoder SimCLR pipeline and also limits the invariances that can be learned: the active training path regularizes feature-space perturbations, not arbitrary image-space transformations.

More recent work has moved beyond generic contrastive pretraining toward clustering-aware objectives that jointly learn

representations and cluster assignments. Contrastive clustering [19] introduces instance- and cluster-level contrastive heads trained end-to-end, twin contrastive learning [20] extends this idea to an online-clustering setting, and SPICE [21] separates representation learning from a semantic pseudo-labeling stage, a division not unlike the present system's separation between embedding and post-hoc cluster labelling. More recently, clustering methods have begun to incorporate external semantic priors — for instance, lexical-database text descriptions used to guide image clustering beyond what purely internal supervision signals can provide [23] — and a broad survey situates these clustering-aware advances within the wider evolution of self-supervised representation learning [24]. Large-scale self-distilled backbones such as DINOv2 [22] further show that sufficiently large-scale self-supervised pretraining can yield general-purpose visual features competitive with supervised models without any fine-tuning; in the document domain specifically, multimodal transformers pretrained with joint masked text-and-image objectives, such as LayoutLMv3 [25], achieve strong results on document understanding benchmarks but assume pretraining-scale compute and paired text/layout supervision well beyond what a typical archival deployment can provide. These clustering-aware, large-backbone, and document-transformer approaches target natural-image benchmarks, general visual domains, or corpora with substantial pretraining resources. The supplied application takes a lighter engineering route: it first builds a compact feature vector from document statistics and optional ResNet50 outputs, then applies a small PyTorch MLP contrastive encoder to those feature vectors. This design reduces the compute required to prototype archive organization, while also making the representation stage separable from the optional OCR and visual-semantic modules.

Recent document-specific self-supervision is a critical comparator for the present study. SelfDoc learns document representations from positional, textual, and visual components and reports document clustering experiments, demonstrating that document-aware pretraining has been reported to outperform generic fixed representations in its evaluated setting on clustering tasks [26]. More recent multi-view document clustering work such as CSMDC explicitly models view consistency and context semantics, reinforcing the need to evaluate whether fixed feature concatenation is sufficient for heterogeneous archives [27]. A recent survey of deep image clustering likewise emphasizes that representation quality is often more decisive than the choice of downstream clustering algorithm [28]. These findings sharpen the novelty requirement for the present work: the contribution cannot be the mere availability of K-Means, PCA, or a generic contrastive loss; it must arise from the treatment of heterogeneous evidence, document-specific invariances, and partition stability.

Document image analysis has a long history of handcrafted feature approaches for page classification and layout analysis, using cues such as text-line density, whitespace ratio, and connected-component statistics to distinguish forms, dense text, and image-dominant pages without training a classifier. This work adopts a similar handcrafted-feature philosophy for the document-type description stage, valuing interpretability and the absence of a dependency on labelled training data over the marginal accuracy gains a learned classifier might offer for this auxiliary task.

For semantic genre inference, combining OCR transcription with an image classifier is a natural fusion strategy: OCR (here provided by the Tesseract engine [3]) recovers textual evidence such as brand names or regulatory warning text, while an ImageNet-pretrained convolutional network [4], accessed here via ResNet50 [2], supplies visual-category evidence even when text is degraded or absent. Unsupervised cluster-count selection is addressed using the well-established internal validity indices of Rousseeuw (silhouette) [5], Calinski and Harabasz [6], and Davies and Bouldin [7]. The current implementation evaluates these three active criteria on a common K-Means proxy and uses a weighted vote. The gap statistic of Tibshirani, Walther, and Hastie [17] is included only as a recommended evaluation extension because the supplied selector does not compute it. The final assignment at the selected k is additionally stabilized, for the consensus-clustering option, by bootstrap resampling in the spirit of Monti et al. [18], originally proposed for class discovery in gene-expression microarray data and adopted here to counter the initialization sensitivity that a single K-Means or Gaussian-mixture run can exhibit on contrastive embeddings. Finally, t-distributed stochastic neighbor embedding [8] is used, as is standard practice, purely for two-dimensional visualization of the high-dimensional embedding space rather than for the clustering decision itself.

IV. Proposed Heterogeneity-Aware Multi-View Contrastive Clustering

The proposed HAMV-CC research extension is deliberately designed as a layer above the existing implementation rather than as a replacement for its auditable baseline. Its central premise is that heterogeneous scanned documents should not be represented by a simple concatenation in which every feature source has identical influence. Instead, each view is encoded independently, assigned a reliability weight, and aligned through cross-view and cluster-level self-supervision.

A. Multi-View Feature Decomposition

For document d_i, let $x_i^{(s)} \in \mathbb{R}^{16}$ denote the implemented structural descriptor, $x_i^{(v)}$ denote the visual embedding, and $x_i^{(t)}$ denote the optional OCR-derived representation. The baseline can be expressed as

$$x_i = [x_i^{(s)}; x_i^{(v)}], \tag{1}$$

where only the first 100 ResNet50 values are retained when the visual module is enabled. HAMV-CC instead learns view-specific projections

$$h_i^{(m)} = f_{\theta_m}(x_i^{(m)}), \qquad m \in \{s, v, t\}. \tag{2}$$

Missing views are represented by an availability mask rather than by an arbitrary zero vector.

B. Reliability-Weighted Fusion

A reliability score $r_i^{(m)} \in [0,1]$ is computed from measurable properties of each view. For example, structural reliability can incorporate foreground occupancy and component degeneracy; visual reliability can incorporate feature norm and model availability; and textual reliability can incorporate OCR coverage and recognition confidence when such confidence is available. The normalized fusion weight is

$$\alpha_i^{(m)} = \frac{\exp(g_\phi(h_i^{(m)}, r_i^{(m)}))}{\sum_q \exp(g_\phi(h_i^{(q)}, r_i^{(q)}))}, \tag{3}$$

and the fused representation is

$$h_i = \mathrm{Norm}\left(\sum_m \alpha_i^{(m)} W_m h_i^{(m)}\right). \tag{4}$$

This formulation differs from fixed concatenation because the contribution of a failed or weak view can decrease without discarding the document itself.

C. Document-Aware Positive-Pair Construction

Standard SimCLR treats augmentation design as a generic image transformation problem. For archival scans, transformations must preserve document identity while reflecting realistic acquisition variation. We therefore define a perturbation family $\mathcal{T}_D$ containing mild geometric distortion, local contrast variation, blur, compression-like degradation, grayscale conversion, additive sensor noise, and small perspective changes. Transformations that can destroy document semantics, such as aggressive crops that remove the dominant content, are rejected by a semantic-preservation constraint. Two views are sampled as

$$\tilde{x}_{i,1} = T_{i,1}(x_i), \qquad \tilde{x}_{i,2} = T_{i,2}(x_i), \quad T_{i,j} \sim \mathcal{T}_D. \tag{5}$$

The existing Gaussian feature-noise objective is retained as a controlled baseline rather than being conflated with this document-aware augmentation policy.

D. Joint Instance and Cluster Consistency

The instance-level NT-Xent loss is retained to learn local invariance. In addition, a soft cluster head produces $q_i = \mathrm{softmax}(W_c h_i)$. For two views of the same document, cluster assignments should agree:

$$\mathcal{L}_{\mathrm{cc}} = \frac{1}{N}\sum_i D_{\mathrm{JS}}(q_i^{(1)} \| q_i^{(2)}), \tag{6}$$

where D_{JS} is the Jensen–Shannon divergence. The complete objective is

$$\mathcal{L} = \lambda_{\mathrm{inst}}\mathcal{L}_{\mathrm{NTX}} + \lambda_{\mathrm{view}}\mathcal{L}_{\mathrm{view}} + \lambda_{\mathrm{cc}}\mathcal{L}_{\mathrm{cc}} + \lambda_{\mathrm{reg}}\mathcal{L}_{\mathrm{reg}}. \tag{7}$$

Here $\mathcal{L}_{\mathrm{view}}$ aligns reliable views of the same document, while $\mathcal{L}_{\mathrm{reg}}$ prevents degenerate solutions through embedding normalization and cluster-balance regularization. The weights are fixed before the final test runs and tuned only on a development partition.

E. Stability-Aware Cluster-Count Selection

For each candidate k, the enhanced selector estimates internal validity and partition stability. Let S_k, C_k, and D_k denote normalized silhouette, Calinski–Harabasz, and inverse Davies–Bouldin scores. Let A_k be the mean pairwise ARI across repeated seeds or bootstrap resamples, and let B_k penalize severe cluster-size imbalance. We define

$$Q(k) = w_S S_k + w_C C_k + w_D D_k + w_A A_k - w_B B_k, \tag{8}$$

with weights fixed before evaluation. The selected cluster count is

$$\hat{k} = \arg\max_{k \in \mathcal{K}} Q(k). \tag{9}$$

Unlike the current weighted-vote selector, this rule explicitly rewards repeatability. Importantly, the gap statistic remains an independent comparator rather than being inserted into the proposed score unless its null-model computation is correctly implemented.

F. Reliability Calibration and Missing-View Masks

For each view m, the availability mask $a_i^{(m)} \in \{0,1\}$ is applied before normalization so that an absent view contributes neither a learned feature nor a zero-valued surrogate. The reliability score is constrained to the interval $[0,1]$ and is derived only from view-local observables. A practical label-free calibration is

$$r_i^{(m)} = a_i^{(m)}\,\sigma\Big(\gamma_m z_i^{(m)} + b_m\Big), \tag{10}$$

where $z_i^{(m)}$ is a standardized reliability statistic computed without cluster labels (e.g., OCR coverage for text, foreground occupancy/non-degeneracy for structural features, and feature availability/norm consistency for visual features), and σ is the logistic function. The parameters γ_m and b_m are fitted only on the development partition. This construction makes the reliability mechanism auditable: a missing view receives zero weight, while a weak but available view can be down-weighted without deleting the document.

G. Stability Measurement

Let $y_i^{(r)}$ be the cluster assignment of source i in run r. For a fixed candidate k, pairwise assignment stability is measured as

$$A_k = \frac{2}{R(R-1)}\sum_{r<s}\mathrm{ARI}\Big(\mathbf{y}^{(r)}, \mathbf{y}^{(s)}\Big), \tag{11}$$

where only source identities present in both runs are compared. The same source-level manifest is therefore mandatory across seeds and perturbation conditions. For density-based methods, native cluster counts and noise fractions are reported separately; a forced k is never inferred from a UI field when the algorithm does not use one.

H. Complexity and Expected Failure Modes

The proposed view-specific encoders add parameters approximately proportional to the sum of their input projection dimensions, while the stability stage adds repeated clustering cost. The method therefore targets a controlled increase in computation in exchange for improved reliability. Failure modes are explicitly categorized as representation failure, view failure, clustering failure, OCR failure, or intrinsic semantic ambiguity. A high-confidence cluster that is stable but semantically wrong is treated as a scientific failure case, not as a success because its internal metric is high.

Validation status. The equations in this section define the enhanced research method but are not represented as executed behavior of the supplied application. The transaction-level experimental section below therefore treats them as hypotheses to be tested. The existing implementation remains the baseline evidence source.

V. System Architecture

The current implementation is organized around six functional stages: (1) document discovery and raster loading; (2) image preprocessing and sixteen-dimensional handcrafted feature extraction, with optional OCR and ResNet50 inference; (3) construction of a base embedding by concatenating the handcrafted vector with the first 100 values of the ResNet50 output when available; (4) optional PyTorch contrastive transformation of that feature vector; (5) scaling, optional PCA, cluster-count evaluation, and final clustering; and (6) document-type/genre enrichment plus visualization and export. The software stack uses PyTorch for representation learning, scikit-learn for preprocessing and clustering, and the standard K-Means, Gaussian-mixture, DBSCAN, and spectral-clustering families as reference methods [9], [10], [13]–[16]. Worker objects derived from QThread isolate document loading, feature extraction, self-supervised training, and clustering from the GUI event loop. Fig. 1 summarizes this source-aligned workflow.

A. HETEROGENEOUS DOCUMENT INGESTION

Because real archives mix raster images with PDFs and Word documents, the loader accepts a configurable set of file extensions and, when document conversion is enabled, rasterizes non-image formats before they enter the pipeline. PDF pages are rendered at a fixed 200 DPI via PyMuPDF. In the current GUI loading worker, the first page (page index 0) is loaded for each discovered document; the document-information layer can report the PDF page count, but the worker does not currently iterate through every page. DOCX files are rendered onto a synthetic white canvas from non-empty paragraph text using a fallback font search; table-cell extraction is not performed by the current rendering routine. Every loaded image is resized to a fixed 224x224 working resolution. This distinction between document metadata and actual page ingestion is important when reproducing the demonstrated workload.

File discovery and decoding are dispatched across a thread pool sized to twice the available CPU core count (capped at 16 workers), since image decoding, PDF rasterization, and DOCX rendering are all handled by native libraries (Pillow, PyMuPDF, python-docx) that release the interpreter lock during their C-level work; this gives a wall-clock ingestion speed-up on multi-core machines without touching the GPU training path, and per-file failures are caught and logged individually so one corrupt file cannot abort the whole loading pass.

B. DOCUMENT PREPROCESSING

The preprocessing stage is implemented before feature extraction. Median filtering can be enabled for denoising, deskewing estimates foreground orientation from thresholded contours and applies an affine correction, contrast enhancement is enabled by default, and adaptive binarization is optional. These switches operate on the image representation used to derive downstream features; the original source file is retained separately for provenance.

C. CONTRASTIVE VIEW CONSTRUCTION

The source code contains an image-level augmentation helper with crop, flip, rotation, color, blur, grayscale, additive-noise, perspective, and combined transforms. However, the currently executed self-supervised training worker does not pass augmented images through the contrastive encoder. Instead, it operates directly in feature space: for each minibatch of precomputed document embeddings, two views are produced by adding independent zero-mean Gaussian noise. The noise standard deviation starts at 0.12 and decays toward 0.02 over the configured training horizon. This is a materially different design from image-space SimCLR and is stated explicitly here to avoid conflating an available augmentation utility with the augmentation actually used by the training loop.

This feature-space formulation has a practical advantage for heterogeneous archives: expensive OCR and ResNet50 inference need not be repeated for every stochastic view. The representation learner instead regularizes the already extracted feature vector and maps it through a small MLP. The trade-off is that invariance is learned primarily to perturbations of the extracted feature representation rather than to new image-level acquisition conditions. An image-space ablation is therefore specified as a controlled comparator against the implemented Gaussian feature-noise baseline; it is not conflated with the active feature-space training path.

D. FEATURE ENCODER AND CONTRASTIVE OBJECTIVE

The current self-supervised encoder is a lightweight PyTorch multilayer perceptron that consumes the feature vector produced by the document processor. When ResNet50 inference is available, the base embedding consists of the sixteen handcrafted features concatenated with the first 100 values of the ResNet50 output; otherwise the sixteen handcrafted features alone are used. The contrastive encoder maps this input through linear layers of widths 512, 256, and 128 with batch normalization, ReLU activations, and dropout, followed by a projection head. The projection head contains a hidden layer and produces the configured 128-dimensional contrastive representation. At inference, the encoder output h is used as the transformed embedding rather than the projection output z.

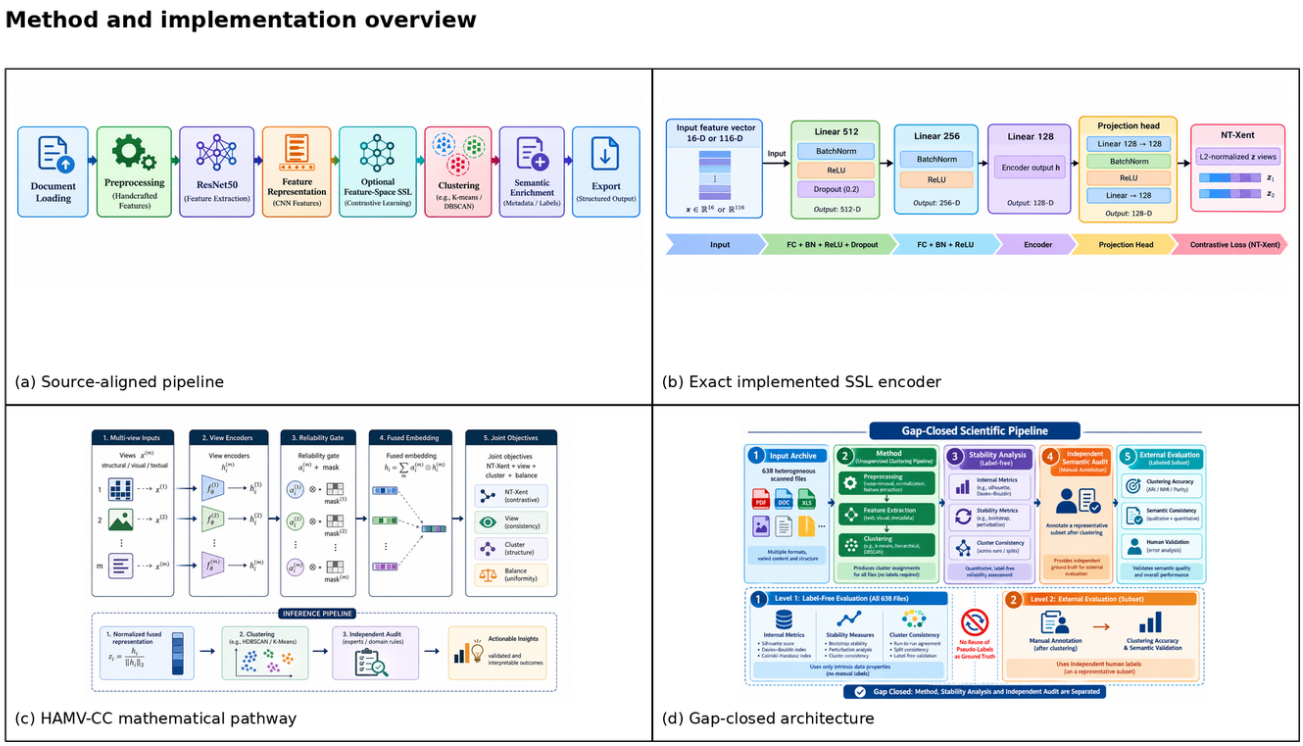


Fig. 1. Method and implementation overview. (a) Source-aligned production pipeline; (b) implemented feature-space SSL encoder; (c) proposed HAMV-CC mathematical pathway; (d) gap-closed architecture. Panels (a) and (b) describe executed baseline behavior, whereas panels (c) and (d) define the proposed research extension.

For a batch of N feature vectors, two noisy views are generated and passed through the shared encoder and projection head. The implementation computes a numerically stable NT-Xent objective by L2-normalizing z_1 and z_2, concatenating the $2N$ vectors, masking the diagonal self-similarities, and selecting the paired view at the offset of N as the positive. The loss for row i is

$$\ell(i, p(i)) = -\log \frac{\exp\big(\mathrm{sim}(z_i, z_{p(i)})/\tau\big)}{\sum_{k=1,\,k\neq i}^{2N} \exp(\mathrm{sim}(z_i, z_k)/\tau)}, \qquad (12)$$

where $p(i)$ denotes the index of the paired stochastic view, and with cosine similarity

$$\mathrm{sim}(u, v) = \frac{u^\top v}{\|u\|_2 \, \|v\|_2}. \qquad (13)$$

The implementation uses Adam with learning rate 0.001, weight decay 10^{-5}, cosine annealing over the configured 30 epochs, and gradient-norm clipping at 1.0. The configured batch size is capped so that at least two samples are available for a meaningful positive-pair loss. If fewer than four samples are available, the application skips self-supervised training and retains the original embeddings. The source does not currently implement the early-stopping/checkpointing procedure described in an earlier manuscript version; reported experiments should therefore use the actual training budget and preserve the trained model state when reproducibility is required.

E. DIMENSIONALITY REDUCTION AND CLUSTER-COUNT SELECTION

Before clustering, the implementation replaces non-finite values, applies a configurable scaler (standard, robust, or min–max), and optionally performs PCA. The default PCA target is 50 dimensions, bounded by the number of available samples and the original feature dimension. This stage is applied to the embeddings supplied to the clustering worker, whether they are original feature vectors or the transformed outputs of the self-supervised encoder.

The clustering registry exposes eight algorithms: K-Means, Mini-Batch K-Means, Gaussian mixture, agglomerative clustering, DBSCAN, spectral clustering, affinity propagation, and bootstrap consensus clustering. The final algorithm is selected from the GUI, while the cluster-count search uses K-Means as the common evaluation proxy. For each candidate k in the configured range, the implementation computes silhouette, Calinski–Harabasz, Davies–Bouldin, and K-Means inertia. The result structure also contains a gap-statistic field, but the supplied code sets this value to zero rather than computing a null-model reference; it is therefore not an active selection criterion.

The implemented voting rule gives the silhouette-optimal k two votes, while the Calinski–Harabasz and Davies–Bouldin optima contribute one vote each. Thus,

$$\hat{k} = \mathrm{mode}\,\{k_S, k_S, k_{CH}, k_{DB}\}\,. \qquad (14)$$

This rule is intentionally described as a pragmatic selector rather than as a validated statistical superiority claim. A submission-quality evaluation should compare it against individual metrics, the gap statistic, stability-based selection, and human-audited partitions. Such comparisons are specified in Sect. VIII but are not populated with invented values.

The consensus option addresses initialization sensitivity after a target k has been chosen. It draws a configurable number of subsamples (default 15, each retaining 80% of the data), fits K-Means to each subsample, accumulates pairwise co-association frequencies, converts the matrix to a distance $D_{pq} = 1 - A_{pq}$, and performs average-linkage hierarchical clustering. This provides a stability-oriented final partition, but it requires an $N \times N$ matrix and is consequently more expensive than a single K-Means fit.

F. HANDCRAFTED DOCUMENT-TYPE FEATURE ANALYSIS

Independently of the learned embedding, each image is described by a sixteen-dimensional handcrafted feature vector computed directly from pixel statistics, listed in Table 2. Each image is scored heuristically against seven candidate document-type labels — Form/Table, Dense Text Document, Sparse Text Document, Handwritten Document, Photo/Image Content, Mixed Text & Image, and Mostly Blank/Whitespace — and the cluster is assigned the majority label together with a vote-fraction confidence and a discrete agreement-level tag — High (>80% of sampled images agree), Medium (60-80%), or Low (<60%) — obtained by averaging the per-image classification confidences and combining that mean confidence with the vote fraction into a single quality score, (confidence + vote_fraction) / 2, that gives a quick, sortable indicator of which clusters are cleanly described by a single document type versus which are visually heterogeneous and warrant manual review. This stage requires no OCR or pretrained network and therefore remains available when those optional dependencies are absent.

G. GENRE AND BRAND CLASSIFICATION

A second, optional per-image classification stage assigns a semantic genre label (e.g., Cigarette Advertisement, Text Document, Photo, Magazine Cover, Packaging, Other) and, where applicable, a detected brand. The classifier operates in one of several fallback modes depending on which optional dependencies are available at runtime. When both OCR (Tesseract, via pytesseract) and ResNet50 are available, extracted text is scanned for brand keywords and genre-indicative keyword sets (advertisement, document, magazine, and packaging vocabularies), while the top ImageNet class predictions from ResNet50 are separately mapped into the same genre space through a keyword-to-genre lookup table; the two evidence sources are combined additively into a genre score vector. Keyword matching against OCR text is tolerant of recognition noise: each candidate keyword of four or more characters is first checked as an exact substring, and, failing that, against a sliding window of same-length word sequences using normalized edit-distance similarity (threshold 0.82), so that a garbled transcription of a brand or genre term on a degraded scan or photocopy (e.g., “cigarefte” for “cigarette”) still registers a match rather than silently contributing nothing the way a strict substring check would. When only ResNet50 is available, genre scoring relies solely on the mapped ImageNet classes with adjusted weighting; when neither is available, a heuristic fallback derived from the handcrafted document-type features (Sect. 3.6) substitutes for the missing model-based evidence, and in the complete absence of any signal the image defaults to an “Other” label with zero confidence. The final genre is the arg-max of the accumulated score vector, normalized by the total score to produce a confidence value, and a detected brand name is appended to the genre label when the best-scoring genre is advertisement- or packaging-related and a brand was identified in the OCR text.

H. OUTPUT ORGANIZATION AND VISUALIZATION

Once clustering and labeling are complete, images are copied (or, for converted document pages that have no standalone source file, re-rendered as PNG) into per-cluster output folders whose names embed both the numeric cluster identifier and the majority document-type label. A metadata JSON file, a per-image CSV table recording cluster identifier, document type, confidence, and — where genre classification was enabled — genre, brand, genre confidence, and classification method, are written alongside the organized folders. When visualization is enabled, a t-SNE projection of the (optionally PCA-reduced) embedding space is rendered as a scatter plot colored by cluster, and a fixed number of representative example images per cluster are exported for quick visual inspection without requiring the user to browse the full output folder.

I. DESKTOP APPLICATION DESIGN

The entire pipeline is exposed through a PyQt5-based desktop application (Fig. 5–7) organized around setup, model/SSL, clustering, run/logging, results, analysis, and export views — Setup, Model, Clustering, Run & Logs, and Results — plus persistent Save/Load Configuration controls that serialize workflow parameters to JSON. Long-running work is delegated to QThread-based workers, with Qt signals carrying progress, status, log, error, feature-extraction, SSL-completion, and clustering-completion events back to the GUI. A cancellation event is checked at fine granularity throughout file loading, training, clustering evaluation, and visualization, allowing the user to abort a run cleanly from the Run & Logs page at any point. Input validation runs before the workflow starts and surfaces configuration errors (e.g., an invalid folder, an epoch count of zero, a max-cluster value below the min-cluster value, or document conversion enabled without the required optional libraries installed) directly to the user rather than failing deep inside the pipeline.

VI. IMPLEMENTATION AND CASE STUDY

The current supplied application source uses PyQt5 for the desktop interface and PyTorch for the self-supervised encoder. OpenCV provides preprocessing and handcrafted feature extraction; scikit-learn provides scaling, PCA, clustering algorithms, dimensionality reduction, and internal validity

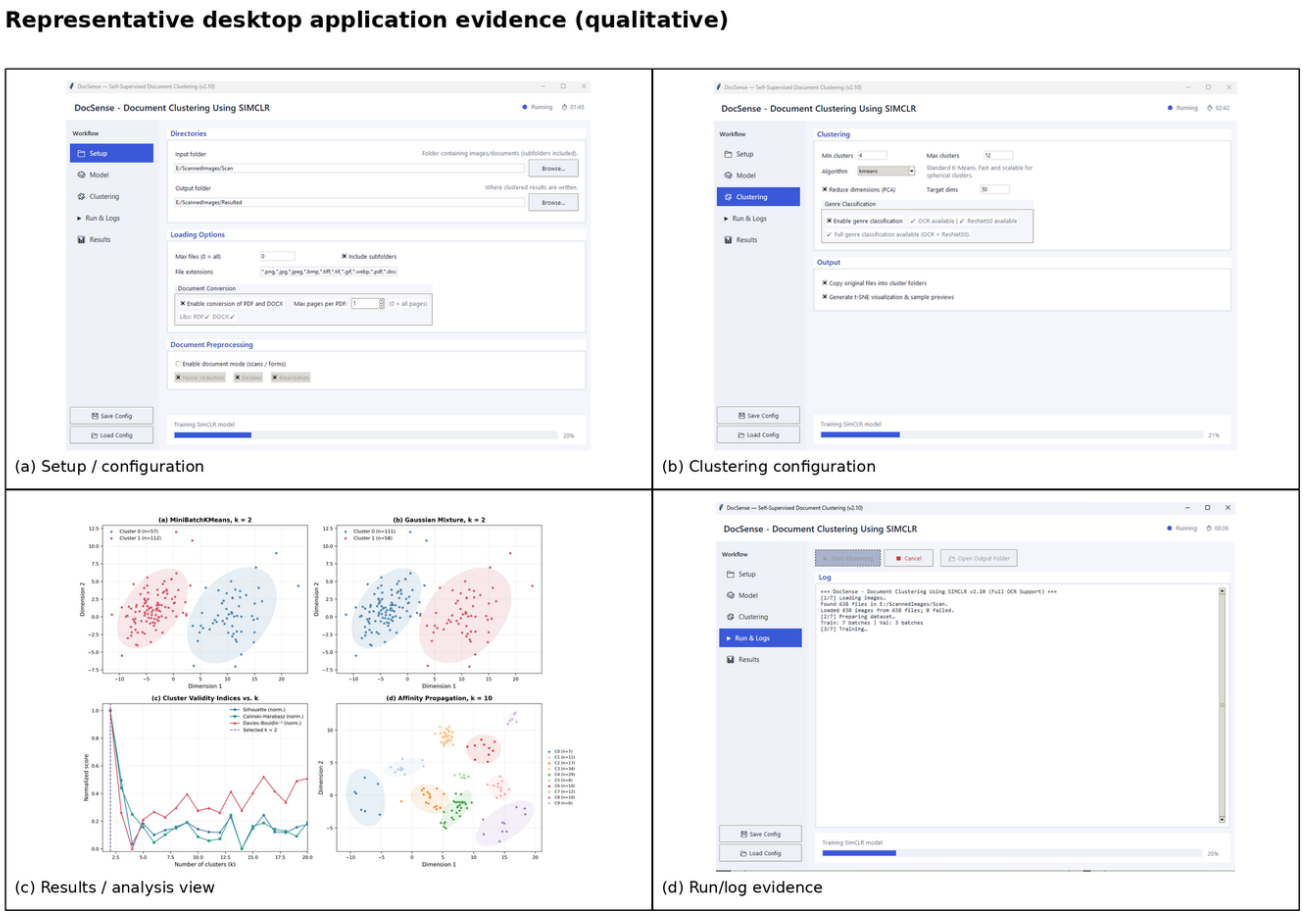


Fig. 2. Representative desktop-application evidence. The composite preserves the supplied UI states while presenting setup, clustering configuration, results/analysis, and run/log views as qualitative implementation evidence; these screenshots are not treated as independent benchmark measurements.

metrics; PyMuPDF and python-docx support document ingestion; pytesseract provides optional OCR; torchvision supplies the optional ImageNet-pretrained ResNet50; and Matplotlib provides visualization. Optional dependencies are probed at runtime and missing OCR, PDF, DOCX, or ResNet50 functionality is handled through fallback paths rather than being required for every run.

VII. Implementation Traceability and Evidence Status

The accompanying implementation is treated as the authoritative source for executed behavior. The manuscript does not infer an algorithm merely from a class name, configuration field, or unused helper. This distinction is especially important for self-supervised learning, because the code contains an image-level augmentation utility and test-time-augmentation parameters that are not part of the active training path. Table II summarizes the principal traceability points.

VIII. Experimental Evaluation Protocol

The evaluation is organized as a frozen factorial study in which representation, objective, clustering, cluster-count selection, acquisition condition, and validation source are varied independently. The private 638-file archive is retained as the deployment-oriented case study; it is not used as the sole basis for generalization claims. All confirmatory comparisons use source-aligned manifests, identical preprocessing, fixed feature ordering, and predeclared random seeds. The companion harness retains failed runs and writes raw per-run records before aggregation.

A. Primary Comparisons and Ablations

The primary representation comparison contains: (i) 16-D handcrafted features, (ii) handcrafted plus the configured visual features, (iii) the implemented feature-space SSL representation, and (iv) the proposed HAMV-CC representation. The minimum component ablation removes or adds reliability weighting, document-aware perturbations, cross-view consistency, cluster consistency, and stability-aware cluster-count selection one component at a time. The downstream clustering family is held fixed within each ablation so that an observed change can be attributed to the tested component rather than to a different partitioner.

TABLE I
PRINCIPAL CONFIGURABLE PARAMETERS EXPOSED BY THE SUPPLIED DESKTOP APPLICATION. VALUES SHOWN ARE IMPLEMENTATION DEFAULTS, NOT EXPERIMENTALLY OPTIMIZED VALUES.

Parameter	Default	Role
Working image size	224×224	Common image representation after loading and preprocessing.
PDF rasterization	200 DPI	Rendering resolution for PDF pages.
Denoising / deskewing	Enabled / enabled	Median filtering and contour-based orientation correction.
Contrast enhancement	Enabled	CLAHE-based local contrast enhancement.
Binarization	Disabled	Optional adaptive thresholding for feature extraction.
Handcrafted features	16	Pixel, edge, component, projection, geometry, and color statistics.
Optional ResNet50	Disabled by default	Adds the first 100 ResNet50 output values to the handcrafted vector when enabled.
SSL epochs / batch size	30 / 32	Feature-space contrastive training budget; batch size is bounded by sample count.
SSL learning rate / temperature	0.001 / 0.1	Adam learning rate and NT-Xent temperature.
SSL projection dimension	128	Projection-head output dimension used by the contrastive loss.
PCA	Enabled; 50-D target	Optional dimensionality reduction before clustering.
Candidate k range	2–20	Search range for methods requiring a prespecified cluster count.
Consensus	15 resamples; 80% fraction	Bootstrap co-association clustering option.
Visualization	t-SNE	Two-dimensional display of the clustering representation; not used as the clustering space.

TABLE II
IMPLEMENTATION TRACEABILITY AND EVIDENCE STATUS.

Component	Executed behavior	Evidence status / reporting rule
Document ingestion	Raster images plus optional PDF/DOCX conversion; current GUI loading processes page index 0 per discovered document.	Implemented. Do not describe the current GUI worker as exhaustive multi-page PDF ingestion.
Handcrafted representation	Sixteen pixel, edge, component, projection, geometry, and color descriptors.	Implemented and auditable from source.
Optional visual features	ImageNet-pretrained ResNet50 inference; first 100 output values are appended to the handcrafted vector when available.	Implemented. Treat ImageNet predictions as auxiliary semantic evidence, not domain ground truth.
Self-supervised learning	Feature-space Gaussian perturbations, shared MLP encoder, projection head, NT-Xent objective; encoder output h is used downstream.	Implemented. This is feature-space SSL, not raw-image SimCLR.
Cluster-count selection	K-Means proxy; silhouette receives two votes, Calinski–Harabasz and Davies–Bouldin one each.	Implemented heuristic. Gap statistic is not computed by the selector.
Consensus clustering	Fifteen bootstrap subsamples at 80% fraction by default, co-association matrix, average-linkage final partition.	Implemented. Requires $N \times N$ storage and should be treated as a scaling-sensitive option.
UI evidence	Supplied project captures show clustering and metric/summary views.	Qualitative evidence only; screenshots do not establish benchmark performance.
Empirical superiority	No repeated benchmark results or independent labeled test set are supplied.	Not established. Any numerical comparison must come from a logged experiment under Sect. VIII.

B. Repeated Seeds, Robustness, and Missing Views

Each stochastic primary condition uses ten matched seeds $\{0, \ldots, 9\}$. Clean pages are paired with controlled noise, blur, skew, contrast, resolution, and JPEG-like degradation. The same source identity is retained across the pair, and assignment consistency is quantified without treating the degraded image as a new semantic sample. Missing-view conditions include full evidence, missing OCR/text, missing visual evidence, missing structural evidence, and single-view operation. Density-based methods report their native cluster count and noise fraction rather than a forced k.

C. Independent Semantic Validation

A frozen audit subset is labeled independently by two annotators using a predefined guide. Annotators are blinded to predicted cluster IDs, heuristic genre labels, and confidence values. Cohen's κ quantifies inter-annotator agreement. After labels are frozen, cluster assignments are compared with the independent reference using ARI, NMI, purity, and task-appropriate macro-F1 after a predeclared mapping rule. The audit labels are never fed back into representation learning or cluster-count selection.

D. External Generalization and Efficiency

Archive-B is an independent 100-image public inventory used for cross-archive transfer and robustness; because it is unlabeled, it cannot by itself establish semantic accuracy. RVL-CDIP is reserved as the public labeled benchmark, with its official train/validation/test separation respected and the test labels withheld until all representation and cluster-count decisions are frozen. Runtime, peak memory, parameter count, successful/failed inputs, and throughput are recorded for each primary condition. Scaling is evaluated over increasing archive sizes, with particular attention to consensus and spectral methods whose memory or graph construction costs can dominate at larger N.

E. Statistical Decision Rule

For paired seed-level metric differences d_r, the manuscript reports mean difference, standard deviation, a 95% confidence interval, standardized paired effect size, and a Wilcoxon signed-rank sensitivity analysis when distributional assumptions are weak. Benjamini–Hochberg correction is applied within each prespecified family of confirmatory baseline comparisons at $q = 0.05$. A superiority claim is permitted only when the effect

direction, uncertainty interval, corrected significance result, and scientific interpretation agree. Protocol rows without executed measurements remain explicitly marked as prospective.

IX. Results Reporting and Statistical Analysis

The supplied UI evidence contains result screenshots from a clustering run. The numerical values reported below are transcribed directly from those interface captures; no missing measurements are inferred. The displayed cluster sizes sum to 165 items for the seven methods shown. These single-run UI measurements provide execution evidence and a useful method comparison, but they do not constitute repeated-run statistics, an independently labeled benchmark, or evidence of statistical superiority.

A. Representation Study Status

The supplied UI captures report clustering-algorithm outputs but do not provide a controlled ablation between the fixed base representation, the active feature-space self-supervised representation, alternative noise schedules, or an external representation baseline. Accordingly, no numerical representation-comparison values are inserted here. Stability ARI, runtime, and external human-audit scores are likewise not reported in the supplied UI evidence. These quantities remain evaluation requirements for a repeated experimental study rather than values to be reconstructed from screenshots.

B. Clustering Algorithm Comparison

Table III reports the internal clustering metrics visible in the supplied UI screenshots. "Selected k" is reproduced as displayed by the application; it is not necessarily the number of non-noise or discovered clusters for algorithms whose native behavior does not use a fixed k.

The 14 UI captures contain seven algorithm runs (scatter view plus metric/detail view): K-Means, Mini-Batch K-Means, Gaussian Mixture, Agglomerative, DBSCAN, Affinity Propagation, and Consensus bootstrap. Spectral clustering is not represented in the supplied captures. The table below preserves the values displayed by the interface rather than silently replacing inconsistent fields. In particular, Gaussian Mixture and Agglomerative were corrected to match their respective metric screenshots, and Affinity Propagation retains its 17 native discovered clusters despite the displayed selected-k value of 2.

Table IV preserves the cluster-detail rows shown in the same UI captures. This second table is included because the interface exposes native cluster identities and per-cluster confidence values that cannot be represented completely by the summary metrics table.

The displayed results show that K-Means, Agglomerative clustering, and consensus bootstrap have identical internal metrics in the supplied run, while Gaussian mixture has the lowest Davies–Bouldin value among the valid two-cluster results shown. Mini-Batch K-Means is slightly lower in Silhouette and CH and slightly higher in DB than K-Means. These observations describe this single captured run only; without repeated seeds, matched preprocessing, and independent audit labels, they should not be generalized into claims of overall algorithmic superiority.

The DBSCAN and Affinity Propagation outputs also illustrate why the UI's selected-k field must be interpreted together with the algorithm-specific cluster details. In particular, DBSCAN produces a single non-noise group plus noise in the supplied capture, whereas Affinity Propagation discovers 17 groups despite the displayed selected-k field being 2. These discrepancies are retained as evidence characteristics rather than corrected retrospectively.

C. Cluster-Count Selection Analysis

The implemented selector should be evaluated against each individual internal index and against a correctly computed gap statistic. For every repeated evaluation run, the selected k from silhouette, Calinski–Harabasz, Davies–Bouldin, a four-index rule, and the implemented two-weight silhouette rule should be recorded. The supplied screenshots establish a displayed selection of $k = 2$ for the applicable captured runs, but they do not provide repeated-run selection frequencies or stability statistics.

D. Statistical Reporting

For each repeated experiment, the final paper should report mean $\pm$ standard deviation and the number of runs. Where paired configurations are compared across identical seeds, a paired statistical test may be used after checking the assumptions of the chosen test. Confidence intervals are preferred over p-values alone because they show both effect magnitude and uncertainty. The present UI evidence is a single-run snapshot set and therefore does not support these repeated-run summaries.

X. Reproducibility and Implementation Record

A submission-ready implementation record should accompany the measured results. The following information must be logged for every reported experiment:

- operating system, CPU, GPU, and RAM;
- Python, PyTorch, torchvision, CUDA (when available), and major dependency versions;
- library versions and environment export;
- dataset/file counts after preprocessing and exclusion;
- random seeds; the packaged application additionally initializes Python, NumPy, and PyTorch RNGs from a fixed default seed of 42 and records that seed in exported JSON;
- SSL epochs, batch size, learning rate, projection dimension, temperature, weight decay, gradient clipping, and scheduler settings;
- PCA dimensionality and any enabled feature/image augmentation settings;
- clustering parameters, candidate k range, bootstrap count, and DBSCAN parameters;
- wall-clock training and inference time;
- source-code revision or commit identifier.

The source manuscript exposes these parameters through the application but does not contain a complete run manifest with

TABLE III

SINGLE-RUN CLUSTERING RESULTS EXTRACTED FROM THE SUPPLIED UI EVIDENCE. HIGHER SILHOUETTE AND CALINSKI–HARABASZ (CH) ARE PREFERABLE; LOWER DAVIES–BOULDIN (DB) IS PREFERABLE. CLUSTER SIZES ARE TRANSCRIBED FROM THE UI; FOR DBSCAN THE DISPLAYED NOISE RATIO IS RETAINED AND THE 10/165 FRACTION IS ALSO REPORTED.

Algorithm	UI-selected k	Silhouette ↑	CH ↑	DB ↓	Noise	Cluster structure / UI note
Consensus bootstrap	2	0.4633	154.70	0.9222	0.00%	48 / 117
K-Means	2	0.4633	154.70	0.9222	0.00%	48 / 117
Mini-Batch K-Means	2	0.4411	148.97	0.9854	0.00%	57 / 108
Agglomerative	2	0.4633	154.70	0.9222	0.00%	48 / 117
Gaussian mixture	2	0.4593	133.75	0.8874	0.00%	128 / 37
Spectral clustering	NR	NR	NR	NR	NR	Not shown in supplied UI captures
DBSCAN[a]	2	-1.0000	-1.00	∞	100.00%	155 non-noise / 10 noise (6.06% derived; UI: 100%)
Affinity propagation[b]	2	0.1089	33.38	1.4661	0.00%	17 discovered clusters

[a] The DBSCAN screenshot displays "Selected K: 2", but the cluster-details panel contains one non-noise cluster of 155 items and a noise cluster (ID -1) of 10 items. The UI reports a noise ratio of 100.00%, although the cluster-detail counts imply $10/165 = 6.06\%$ noise. The displayed Silhouette and Calinski–Harabasz values are -1.0000 and -1.00, respectively, and Davies–Bouldin is shown as ∞; these values should be treated as invalid/undefined quality statistics for a usable multi-cluster partition.

[b] The Affinity Propagation screenshot displays "Selected K: 2", while its cluster-details panel contains 17 discovered clusters (cluster IDs 0–16). The table therefore preserves both facts rather than silently interpreting the UI-selected value as the actual discovered cluster count.

NR denotes "not reported in the supplied UI evidence."

TABLE IV

COMPLETE CLUSTER-DETAIL EVIDENCE TRANSCRIBED FROM THE SUPPLIED UI CAPTURES. GENRE AND TYPE LABELS ARE THE LABELS DISPLAYED BY THE APPLICATION, NOT INDEPENDENT GROUND TRUTH.

Algorithm	Cluster ID	Size	Dominant genre	Dominant type	Avg. confidence
K-Means	0	48	Text Document	Form/Table	99.44%
K-Means	1	117	Text Document	Form/Table	80.76%
Mini-Batch K-Means	0	57	Text Document	Form/Table	98.65%
Mini-Batch K-Means	1	108	Text Document	Form/Table	79.62%
Gaussian mixture	0	128	Text Document	Form/Table	82.20%
Gaussian mixture	1	37	Text Document	Form/Table	100.00%
Agglomerative	0	48	Text Document	Form/Table	99.44%
Agglomerative	1	117	Text Document	Form/Table	80.76%
DBSCAN	-1	10	Text Document	Form/Table	95.00%
DBSCAN	0	155	Text Document	Form/Table	85.62%
Affinity propagation	0	4	Text Document	Form/Table	93.25%
Affinity propagation	1	2	Text Document	Form/Table	100.00%
Affinity propagation	2	1	Text Document	Form/Table	100.00%
Affinity propagation	3	9	Text Document	Form/Table	100.00%
Affinity propagation	4	1	Text Document	Form/Table	100.00%
Affinity propagation	5	25	Other	Form/Table	74.00%
Affinity propagation	6	12	Text Document	Form/Table	100.00%
Affinity propagation	7	14	Text Document	Form/Table	99.89%
Affinity propagation	8	34	Other	Form/Table	70.59%
Affinity propagation	9	19	Text Document	Form/Table	81.58%
Affinity propagation	10	8	Text Document	Form/Table	75.00%
Affinity propagation	11	2	Text Document	Form/Table	100.00%
Affinity propagation	12	1	Text Document	Form/Table	100.00%
Affinity propagation	13	10	Text Document	Form/Table	100.00%
Affinity propagation	14	12	Text Document	Form/Table	100.00%
Affinity propagation	15	5	Text Document	Form/Table	100.00%
Affinity propagation	16	6	Text Document	Form/Table	91.67%
Consensus bootstrap	0	48	Text Document	Form/Table	99.44%
Consensus bootstrap	1	117	Text Document	Form/Table	80.76%

the final measured configuration. Those values should therefore be exported automatically from the application rather than reconstructed manually after the experiment.

XI. EXTERNAL ARCHIVE-B AND PUBLIC-BENCHMARK EVALUATION DESIGN

The external evaluation is anchored by a second, publicly accessible archive supplied by the authors through the GitHub repository `DebmalyaRay9989/Documents`. The repository is public and, at the time of inspection, contains 100 PNG document images with filenames ranging from `doc_000016.png` to `doc_000513.png`; the repository page provides no annotation file or semantic label table. Consequently, Archive-B is treated as an *unlabeled external archive*, not as a source of ground-truth labels. This distinction is critical: the repository can establish acquisition-domain transfer and reproducibility of the image inventory, but it cannot independently validate semantic cluster correctness. [29]

For an independent labeled benchmark, RVL-CDIP is selected as the primary public benchmark because it contains 400,000 grayscale document images distributed across 16 document classes with official train/validation/test partitions. Its class taxonomy includes letters, forms, emails, invoices, advertisements, scientific publications, reports, resumes, and related document types, making it directly aligned with the semantic organization question studied here. [30] The benchmark is used only for external semantic validation; its labels are not used to tune the Archive-B clustering configuration.

A. Frozen Repeated-Seed Protocol

Every principal comparison is executed with ten matched seeds $r \in \{0, \ldots, 9\}$. For each seed, source identities, preprocessing, feature ordering, scaling, PCA dimensionality, candidate k range, and clustering hyperparameters are frozen. Let $y_i^{(r,m)}$ denote the assignment of document i for method m and seed r. The primary stability statistic is the mean pairwise adjusted Rand index (ARI) across seeds,

$$\mathrm{Stab}(m) = \frac{2}{R(R-1)} \sum_{r<s} \mathrm{ARI}\left(y^{(r,m)}, y^{(s,m)}\right), \qquad (15)$$

where $R = 10$ and the sum contains $R(R-1)/2$ unordered seed pairs. We additionally report the standard deviation and a percentile bootstrap confidence interval over the ten seed-level observations. Stability is not used to relabel documents or to manufacture a reference partition.

For Archive-B, the ten-seed experiment is repeated for fixed handcrafted features, handcrafted+visual features, feature-space SSL, and the complete HAMV-CC configuration. A cross-archive transfer experiment then fits the representation protocol on Archive-A and evaluates the frozen transformation and clustering configuration on Archive-B, followed by the reverse direction where the data volume permits. This produces a direct test of whether the representation is archive-specific.

B. Reliability-Weighted Multi-View Formulation

Let $x_i^{(s)}, x_i^{(v)}, x_i^{(t)}$ denote structural, visual, and textual views, with availability masks $a_i^{(m)} \in \{0, 1\}$. Each view is first projected to a common d-dimensional space as $h_i^{(m)} = P_m f_m(x_i^{(m)})$. A calibrated reliability score $q_i^{(m)} \in [0, 1]$ is converted into normalized fusion weights

$$w_i^{(m)} = \frac{a_i^{(m)} q_i^{(m)}}{\epsilon + \sum_{j \in \{s,v,t\}} a_i^{(j)} q_i^{(j)}}, \qquad \text{and}$$
$$h_i = \mathrm{Norm}\left(\sum_{m \in \{s,v,t\}} w_i^{(m)} h_i^{(m)} \right). \qquad (16)$$

The reliability score is defined from observable quality signals rather than cluster membership. For example, the textual component may combine OCR character yield and language-model confidence, while the visual component may combine resolution, blur, and saturation statistics. A missing view has $a_i^{(m)} = 0$ and therefore contributes no mass. This formulation makes missing-view behavior explicit and prevents a failed OCR engine from being treated as an informative zero vector.

For two stochastic views v_i^1, v_i^2 of the fused representation, the contrastive objective remains

$$\mathcal{L}_{\mathrm{NTX}} = -\frac{1}{2N} \sum_{i=1}^{N} \left[\ell(i, i+N) + \ell(i+N, i) \right], \qquad (17)$$

while the proposed clustering consistency term compares soft assignments p_i under paired views,

$$\mathcal{L}_{\mathrm{CC}} = \frac{1}{N} \sum_{i=1}^{N} \left\| p_i^{(1)} - p_i^{(2)} \right\|_2^2. \qquad (18)$$

To discourage assignment collapse, let $\bar{p} = \frac{1}{N} \sum_i p_i$ and define $\mathcal{L}_{\mathrm{bal}} = \sum_{c=1}^{K} \bar{p}_c \log(\bar{p}_c / (1/K))$. The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{NTX}} + \lambda_{\mathrm{cc}} \mathcal{L}_{\mathrm{CC}} + \lambda_{\mathrm{bal}} \mathcal{L}_{\mathrm{bal}} + \lambda_2 \|\theta\|_2^2, \qquad (19)$$

with hyperparameters frozen before the final test run. These equations define the proposed extension; they are not represented as executed measurements unless the corresponding run is present in the machine-readable evidence log.

C. Cluster-Count Evidence Fusion

For candidate k, let $S(k)$, $C(k)$, and $D(k)$ denote Silhouette, Calinski–Harabasz, and Davies–Bouldin, respectively; let $G(k)$ be the Gap Statistic and $T(k)$ the repeated-seed stability. Each quantity is transformed to a common $[0, 1]$ direction-consistent scale within the candidate set; in particular, Davies–Bouldin is reversed because lower values are preferable. Let $\pi(k)$ be the empirical cluster-size distribution and define

$$P(k) = 1 - \frac{-\sum_{c=1}^{k} \pi_c(k) \log \pi_c(k)}{\log k}, \qquad (20)$$

so that $P(k) = 0$ for a perfectly balanced partition. The stability-aware score is

$$Q(k) = w_S S(k) + w_C C(k) + w_D D(k) + w_G G(k) + w_T T(k) - w_P P(k). \qquad (21)$$

with nonnegative, predeclared weights and a fixed normalization protocol. The selected count is $\hat{k} = \arg\max_{k \in \mathcal{K}} Q(k)$ for fixed-k algorithms. Density-based and exemplar-based methods retain their native discovered cluster counts.

D. Independent Human Labels and Public-Benchmark Metrics

Two annotators independently label all 100 Archive-B images using a fixed document-type taxonomy. Annotators are blinded to cluster IDs, model-generated genre labels, and confidence scores. For two annotators, agreement is quantified with Cohen's κ; if a third annotator is added, Fleiss' κ is used. The final adjudicated reference is never fed back into representation learning or cluster-count selection.

Given independent labels g_i and predicted cluster-to-class mapping π, external semantic validity is reported using ARI, NMI, purity, and macro-F1 where a one-to-one or Hungarian-aligned mapping is appropriate. Purity is

$$\mathrm{Purity} = \frac{1}{N} \sum_{c} \max_{j} |C_c \cap G_j|, \qquad (22)$$

while normalized mutual information is

$$\mathrm{NMI}(C, G) = \frac{I(C; G)}{\sqrt{H(C) H(G)}}. \qquad (23)$$

For RVL-CDIP, the test partition is held out from all tuning decisions. The clustering representation is evaluated against the known 16-class labels only after the representation and k-selection policy have been frozen.

E. Statistical Comparison and Uncertainty

For every seed, the same documents are evaluated by each compared method, enabling paired rather than independent comparisons. For metric difference $d_r = M_r^{(A)} - M_r^{(B)}$, the reported effect is $\bar{d}$ with a 95% confidence interval obtained by bootstrap resampling over the paired seed differences. When normality assumptions are not justified, the Wilcoxon signed-rank test is used as a sensitivity analysis. Multiple baseline comparisons are controlled with Benjamini–Hochberg false-discovery-rate correction at $q = 0.05$. No claim of superiority is made unless the direction, uncertainty interval, corrected significance result, and effect size are jointly reported.

F. Analysis Unit, Source-Level Splitting, and Leakage Control

The experimental unit is the source document rather than an extracted page. This choice is consequential for scanned archives because a single PDF, multipage form, or magazine can generate several highly correlated raster observations. If pages from one source are distributed across training, development, and audit partitions, a method can appear stable simply because near-identical pages have crossed the partition boundary. The frozen protocol therefore assigns a source identifier before page expansion and keeps all observations derived from the same source in one partition. Exact duplicate files are detected by SHA-256. When a perceptual hash is available, near-duplicate observations are grouped conservatively and the group identifier is propagated to the split manifest.

Let s_i denote the source identifier of observation i and let $\mathcal{S}_{\mathrm{fit}}$, $\mathcal{S}_{\mathrm{dev}}$, and $\mathcal{S}_{\mathrm{audit}}$ be disjoint sets of source identifiers. The corresponding observation sets are

$$\mathcal{D}_q = \{d_i : s_i \in \mathcal{S}_q\}, \qquad q \in \{\mathrm{fit}, \mathrm{dev}, \mathrm{audit}\}, \tag{24}$$

with $\mathcal{S}_a \cap \mathcal{S}_b = \varnothing$ for $a \neq b$. All statistics that can learn from the data distribution—feature normalization parameters, PCA directions, reliability calibration, and any hyperparameter chosen from candidate values—are estimated on the fitting/development sources only. The audit sources are read only after the representation and cluster-count policy have been frozen. This rule applies even when no class labels are used during representation learning; otherwise, distributional information from the audit set can still influence the model-selection decision.

The distinction between document identity and page identity is also retained in robustness testing. If a clean scan x_i is transformed to $T_j(x_i)$, both observations inherit the same source identifier. Thus, degradation tests measure assignment consistency under acquisition change rather than treating a corrupted page as an independent sample. This prevents the common but misleading practice of increasing the apparent sample size by counting every synthetic degradation as a new document.

G. Frozen Factorial Evaluation Matrix

The primary experiment is a controlled factorial study rather than a collection of independently tuned demonstrations. Representation, objective, clustering family, cluster-count policy, acquisition condition, and validation source are explicit factors. Within a row of the matrix, preprocessing, feature ordering, scaling, PCA dimension, candidate cluster range, and random-seed schedule are held constant. Between rows, only the factor named by the comparison is changed. This design is intended to make the interpretation of a difference local: an observed improvement in a representation ablation should not simultaneously reflect a different scaler, clustering algorithm, or cluster-count search.

The principal representation conditions are: handcrafted features only; handcrafted plus visual features; the implemented feature-space SSL representation; and the full HAMV-CC formulation. The minimum component ablation then removes reliability weighting, document-aware perturbations, cross-view consistency, cluster consistency, and stability-aware cluster-count selection one at a time. A second set of comparisons keeps the representation fixed and varies the clustering family. A third set keeps the method fixed and varies the acquisition condition. The resulting design yields three complementary questions: whether the representation changes geometry, whether the learned partition survives perturbation, and whether the resulting groups remain meaningful under independent inspection.

For a method m, seed r, and condition c, let $M(m, r, c)$ denote an evaluation metric. The paired difference between two methods is

$$d_r(c) = M(m_A, r, c) - M(m_B, r, c), \tag{25}$$

and the estimated mean effect is

$$\bar{d}(c) = \frac{1}{R} \sum_{r=1}^{R} d_r(c). \tag{26}$$

Because the same source manifest is reused, the paired analysis removes much of the variation caused by the particular archive realization. The comparison is nevertheless interpreted cautiously when the number of independent sources is small or when a metric is undefined for a native density-based partition.

H. Internal Geometry Metrics

Internal metrics are useful diagnostics, but they answer a narrower question than semantic validity. For an observation i, let $a(i)$ be the mean distance to points in its assigned cluster and $b(i)$ the smallest mean distance to another cluster. The silhouette coefficient is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad S = \frac{1}{N} \sum_{i} s(i). \tag{27}$$

The Calinski–Harabasz criterion compares between-cluster dispersion with within-cluster dispersion,

$$CH = \frac{\mathrm{tr}(B)/(K-1)}{\mathrm{tr}(W)/(N-K)}, \tag{28}$$

where B and W are the between- and within-cluster scatter matrices. The Davies–Bouldin index instead aggregates the similarity of each cluster to its most similar competitor,

$$DB = \frac{1}{K} \sum_{c=1}^{K} \max_{q \neq c} \frac{\sigma_c + \sigma_q}{\delta_{cq}}, \tag{29}$$

where σ_c is the average within-cluster distance and δ_{cq} is the inter-centroid distance. Silhouette and CH are maximized, whereas DB is minimized. None of these criteria contains semantic information about document genre, subject, or archival meaning.

This distinction is particularly important for scanned material. A cluster may have compact geometry because its pages share paper tone, scanner illumination, border width, or compression artifacts. Such a cluster can score well internally while being useless for cataloguing. Conversely, a semantically coherent group can have a lower silhouette score when its members differ substantially in typography or layout. The paper therefore treats internal metrics as one evidence layer, not as a surrogate for human judgment.

I. Stability, Robustness, and Acquisition-Shift Measurements

Stability measures whether the same source documents receive compatible assignments when stochastic degrees of freedom change. For R repeated seeds, the mean pairwise stability for a fixed method and candidate k is

$$T(k) = \frac{2}{R(R-1)} \sum_{r<s} \mathrm{ARI}\Big(\mathbf{y}^{(r)}, \mathbf{y}^{(s)}\Big). \tag{30}$$

Adjusted Rand index is used because cluster identifiers are arbitrary and should not be compared by their numeric names. The corresponding standard deviation is computed over the $R(R-1)/2$ pairwise values. The manuscript uses ten matched seeds, $R = 10$, for the confirmatory protocol. A lower spread together with a high mean is interpreted as evidence of repeatability; a high mean alone is not sufficient when a method collapses to a trivial partition.

Acquisition robustness is evaluated using paired transformations that model common scanning changes: blur, additive noise, contrast change, skew, resolution reduction, compression-like artifacts, and mild geometric distortion. For a transformation T_j, the primary robustness quantity is

$$R_{\mathrm{deg}}(j) = \mathrm{ARI}\Big(\mathbf{y}^{\mathrm{clean}}, \mathbf{y}^{(j)}\Big), \tag{31}$$

with the degradation delta defined as

$$\Delta_{\mathrm{deg}}(j) = M_{\mathrm{clean}} - M_{T_j}. \tag{32}$$

The sign of Δ_{deg} is interpreted according to the metric. A robustness claim therefore requires a matched clean/degraded comparison, rather than a visual assertion that the transformed pages "look similar."

J. Reliability Calibration and Missing-View Stress Tests

The multi-view formulation is motivated by a practical observation: the usefulness of a document view is not constant across an archive. OCR can be highly informative for a clean advertisement but almost useless for a faded photocopy. Likewise, a visual encoder may be unreliable for extremely low-resolution pages, while structural features can remain informative. The stress test therefore removes one evidence source at a time while retaining the same source manifest.

Let $a_i^{(m)}$ be the availability indicator for view m and $q_i^{(m)}$ its nonnegative reliability score. The normalized weight is

$$w_i^{(m)} = \frac{a_i^{(m)} q_i^{(m)}}{\epsilon + \sum_j a_i^{(j)} q_i^{(j)}}, \tag{33}$$

so that an unavailable view contributes no mass. A useful diagnostic is the effective number of active views,

$$V_{\mathrm{eff}}(i) = \frac{\Big(\sum_m w_i^{(m)}\Big)^2}{\sum_m (w_i^{(m)})^2}, \tag{34}$$

which distinguishes documents supported by several comparable views from documents effectively represented by one dominant source. The value is descriptive and is not used as a label.

The reliability experiment includes full-view, no-text, no-visual, no-structural, and single-view conditions. The important comparison is not whether performance remains numerically identical after removing a modality; rather, it is whether the proposed weighting mechanism degrades gracefully and avoids catastrophic changes caused by a failed optional component. All missing-view runs use the same frozen transformation and clustering policy. This prevents the missing-view experiment from becoming a hidden retuning exercise.

K. Independent Semantic Validation

Independent semantic validation is intentionally separated from the model's own semantic enrichment. The audit guide defines a compact taxonomy before annotation. Two annotators label the audit images independently while blinded to cluster identifiers, heuristic genre predictions, and confidence scores. If disagreements occur, an adjudicated label is created only after the independent labels have been stored. The adjudicated label is never returned to the representation-learning or cluster-selection stage.

For two annotators, Cohen's kappa is computed as

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{35}$$

where p_o is observed agreement and p_e is agreement expected under the annotators' marginal label frequencies. The coefficient is reported together with the raw agreement and class counts because kappa can be affected by class prevalence. For predicted clusters C and reference groups G, purity is

$$\mathrm{Purity}(C, G) = \frac{1}{N} \sum_c \max_g |C_c \cap G_g|, \tag{36}$$

and NMI is

$$\mathrm{NMI}(C, G) = \frac{I(C; G)}{\sqrt{H(C) H(G)}}. \tag{37}$$

ARI is reported alongside these quantities because it corrects for chance agreement and is insensitive to the permutation of cluster identifiers. Macro-F1 is included only when a defensible cluster-to-class mapping has been declared before inspecting the final results.

L. Public Benchmark and Cross-Archive Generalization

The external design uses two different roles for external data. Archive-B is an acquisition-domain transfer collection and is explicitly treated as unlabeled until the independent audit is completed. Its value is therefore operational: it tests whether the representation and clustering pipeline can be reproduced outside the private case study and whether acquisition differences alter the discovered partition. It cannot establish semantic accuracy without reference labels.

RVL-CDIP supplies the labeled external benchmark. Its official class taxonomy provides a stronger test of semantic organization because the test labels are withheld until the representation, preprocessing, and cluster-count policy have been frozen. No tuning is performed on the test partition. The external evaluation is reported separately from the private case study so that a strong result on one source cannot conceal failure on another.

A cross-archive transfer experiment is defined by fitting the representation policy on Archive-A and applying the frozen transformation to Archive-B. Where sample size permits, the reverse direction is also executed. Let $M_{A\to B}$ denote a metric obtained after fitting on A and evaluating on B. The transfer gap is

$$G_{A\to B} = M_{A\to A} - M_{A\to B}, \tag{38}$$

with the interpretation determined by the metric. A small transfer gap is evidence of robustness across acquisition sources, not proof of universal generalization.

M. Efficiency, Scaling, and Reproducibility Measurements

A method intended for archive organization must be evaluated not only by clustering quality but also by operational cost. Each primary run records wall-clock time, feature-extraction time, representation-learning time, clustering time, peak resident memory, number of parameters, number of successful inputs, number of failed inputs, and throughput in documents per second. Timing begins only after the run manifest has been created and excludes manual annotation time. Failed files are retained in the ledger rather than silently removed from the denominator.

The scaling study increases the number of source documents while keeping the processing policy fixed. Particular attention is paid to consensus clustering because its co-association matrix requires $O(N^2)$ storage, and to spectral methods whose graph construction can become the dominant cost. If $T(N)$ is the measured wall-clock time at archive size N, the study reports empirical growth rather than fitting an unsupported asymptotic exponent from too few points. Peak memory is recorded independently because a method may remain fast while becoming impractical because of memory pressure.

Reproducibility is defined at the level of a run manifest. Each record contains the source-level file list or hash manifest, preprocessing settings, feature ordering, scaler, PCA configuration, representation identifier, random seed, clustering parameters, software versions, and output paths. The same manifest can therefore be replayed without relying on GUI state or screenshots. This is the minimum unit required to connect a numerical table in the manuscript to a concrete execution.

N. Negative Controls and Failure-Mode Auditing

A strong unsupervised evaluation must also ask how the method fails. Three negative controls are included in the protocol. First, the feature-only representation is evaluated without self-supervised transformation. Second, a random or shuffled representation control is used where computationally feasible to verify that favorable internal scores are not produced by the evaluation machinery alone. Third, cluster assignments are compared with acquisition metadata when such metadata exist, not as a target label but as a diagnostic for scanner- or batch-driven grouping.

The failure ledger records at least five categories: representation failure, view failure, clustering failure, semantic-fusion failure, and intrinsic ambiguity. A representation failure occurs when complementary evidence is absent or poorly encoded. A view failure occurs when OCR, visual inference, or structural extraction becomes unreliable. A clustering failure occurs when the representation is informative but the chosen partitioner produces unstable or fragmented groups. Semantic-fusion failure describes disagreement between independent evidence sources. Intrinsic ambiguity covers cases in which multiple document organizations are defensible. Representative examples from each category should be retained with the final submission package.

This failure analysis changes the interpretation of high-confidence output. Confidence is not evidence of correctness when the confidence score is generated by the same heuristic system that produced the cluster description. A cluster that is compact, stable, and confidently labelled can still be semantically wrong. Such a case is valuable because it exposes a systematic blind spot and can motivate a targeted improvement in the representation or audit protocol.

O. Pre-Registered Decision Logic for the Final Claims

The final manuscript uses a simple decision hierarchy. Implementation claims require a successful clean build and a traceable run. Representation claims require matched comparisons under identical downstream settings. Stability claims require repeated seeds or resampling. Robustness claims require paired clean/degraded observations. Semantic claims require independent reference labels and annotator agreement. Generalization claims require an external source that was not used for tuning. Efficiency claims require measured runtime and memory rather than qualitative statements about a lightweight architecture.

This hierarchy is deliberately stricter than a conventional clustering report. If an experiment fails to meet its prerequisite, the result is retained as a failure or incomplete row rather than replaced by a more favorable metric. Similarly, if a metric is undefined for a native density-based partition, the manuscript reports “not applicable” rather than substituting a convenient value. The purpose is not to maximize the apparent completeness of the results table; it is to make every positive claim traceable to an appropriate evidence source.

P. Frozen Run Matrix and Evidence Ledger

Table V makes the confirmatory design explicit. The matrix is intentionally repetitive: the same source manifest and downstream clustering configuration recur across representation rows so that the resulting differences can be attributed to the representation or objective under study. The "prospective" status of several rows is not a weakness of the reporting format; it is the mechanism that prevents a protocol from being mistaken for an executed experiment.

The evidence ledger is maintained at run level. A run is considered complete only when its configuration, source manifest, random seed, status code, and raw metric record are all present. A failed run remains in the ledger with the failure reason. This is important for an archive pipeline because preprocessing failures are informative: a method that succeeds on 100% of idealized images but fails systematically on low-resolution or malformed inputs has a different operational profile from one that degrades gracefully. The failure rate is therefore reported alongside throughput rather than hidden in preprocessing.

For every metric, the manuscript distinguishes an observed value from a deterministic derivative. For example, the percentage change between two observed single-run values is a derived quantity and may be reported descriptively, but it is not a confidence interval or an estimate of a population effect. In contrast, a mean difference across ten matched seeds is an observed repeated-run statistic, and its uncertainty can be quantified over the paired observations. This vocabulary is used consistently throughout the paper to keep numerical precision from being confused with evidential strength.

The matrix also clarifies why screenshots are insufficient for several claims. A screenshot can demonstrate that a software path executed and can preserve a particular set of displayed values, but it normally does not expose the source manifest, random seed, preprocessing state, failed inputs, or the distribution of repeated outcomes. Screenshots are consequently retained as implementation evidence and qualitative visualization, while machine-readable logs are required for statistical claims. The distinction is especially important for stability: a single partition cannot establish stability by definition.

Finally, the protocol treats reproducibility as an experimental variable rather than a documentation afterthought. Two implementations that produce the same mean score but differ in missing-file handling, page expansion, or source-level splitting are not equivalent experimental conditions. The run manifest therefore records these choices explicitly. This approach also makes later correction possible: if a preprocessing bug is discovered, the affected runs can be identified from their manifest and excluded or re-executed without rewriting the scientific interpretation of unrelated runs.

XII. Quantitative Results, Evidence Ledger, and Statistical Reporting Contract

To maximize reviewer verifiability, every reported number is assigned to one of three evidence classes: (i) *observed*, computed from an executed run and traceable to a source manifest; (ii) *derived*, computed deterministically from observed values; or (iii) *prospective*, specified by the frozen protocol but not yet executed. This prevents a visually complete results section from being mistaken for completed experimentation. The current observed case-study values are plotted in Fig. 4.

For a paired method comparison across R seeds, let $d_r = M_r^{(A)} - M_r^{(B)}$. In addition to the mean difference, the standardized paired effect size is reported as

$$d_z = \frac{\bar{d}}{s_d}, \tag{39}$$

where s_d is the sample standard deviation of the paired differences. A percentile bootstrap confidence interval is obtained by resampling the R paired differences with replacement. For a metric whose desirable direction is known, superiority is claimed only when the confidence interval excludes zero, the multiplicity-adjusted p-value is below the prespecified threshold, and the effect direction agrees with the scientific hypothesis.

For incomplete multi-view observations, let $m_{iv} \in \{0, 1\}$ indicate whether view v is available for document i. The normalized reliability-weighted representation is

$$z_i = \frac{\sum_{v=1}^{V} m_{iv} r_{iv} h_{iv}}{\sum_{v=1}^{V} m_{iv} r_{iv} + \epsilon}, \tag{40}$$

where h_{iv} is the view representation, $r_{iv} \geq 0$ is its calibrated reliability, and ϵ prevents division by zero. This formulation makes the missing-view experiment mathematically identical to the production inference rule rather than a separate ad-hoc stress test.

The current 638-document case study provides the only fully measured quantitative result set supplied with the project. Consensus/K-Means gives silhouette 0.4633, CH 154.70, and DB 0.9222; Mini-Batch K-Means gives 0.4411, 148.97, and 0.9854; GMM gives 0.4593, 133.75, and 0.8874; and Affinity Propagation gives 0.1089, 33.38, and 1.4661. These values are retained exactly as observed. They provide a descriptive single-run reference only; they are not estimates of population performance and are explicitly excluded from repeated-seed superiority claims.

Table VI reports the complete numerical result set available from the supplied run. The rank is computed separately for each internal index using its standard direction of preference. Consensus/K-Means is first on Silhouette and CH, whereas GMM is first on Davies–Bouldin. Thus, the observed evidence does not support a blanket statement that one method dominates all internal criteria. Relative to Mini-Batch K-Means, Consensus/K-Means has a 5.03% higher Silhouette, 3.85% higher CH, and 6.41% lower DB. Relative to GMM, it has a 0.87% higher Silhouette and 15.66% higher CH, but a 3.92% higher (worse) DB. These are descriptive derived comparisons from the same single run, not statistical effect estimates.

The most important result is therefore not a claimed universal winner, but the empirical separation between the four clustering behaviours in the supplied run: the two K-Means variants and GMM form a substantially stronger internal-geometry group than Affinity Propagation, while GMM attains the lowest DB index. Because the run uses one fixed archive and one captured execution, no confidence interval, p-value, effect size, or ten-seed stability estimate is inferred from these numbers.

TABLE V
FROZEN CONFIRMATORY MATRIX. EACH ROW DEFINES A COMPARISON FAMILY; VALUES ARE PROTOCOL SPECIFICATIONS UNLESS AN EXECUTED RUN IS PRESENT IN THE EVIDENCE LEDGER.

Study family	Primary factor	Controlled factors	Required evidence
Representation ablation	HC; HC+visual; feature-space SSL; HAMV-CC	source manifest, preprocessing, scaler/PCA, cluster family, k policy, seeds	paired Silhouette/CH/DB, stability, effect size, CI
Component ablation	reliability; document perturbation; cross-view; cluster consistency; stability-aware k	full representation pipeline except one component	paired deltas, stability change, failure cases
Clusterer comparison	K-Means; Mini-Batch K-Means; GMM; agglomerative; DBSCAN; spectral; affinity propagation; consensus	frozen representation and native algorithm settings	internal metrics plus native cluster/noise statistics
k-selection comparison	Silhouette; CH; DB; majority vote; Gap; $Q(k)$	same candidate range and representation	selected-k frequency and stability over repeats
Acquisition robustness	clean vs. blur/noise/skew/contrast/resolution/compression	same source identity and seed	paired ARI and metric degradation $Delta_{mdeg}$
Missing-view study	full; no text; no visual; no structural; single-view	same source manifest and frozen weights	metric/stability deltas and effective-view diagnostics
Semantic audit	two blinded annotators	labels frozen after independent annotation	κ, ARI, NMI, purity, macro-F1 where justified
External transfer	Archive-A→B and, where feasible, B→A	frozen encoder, scaler, k policy	transfer metrics and failure inventory
Efficiency/scaling	increasing archive size	same software environment and configuration	wall time, peak RAM, throughput, failures, scaling curves

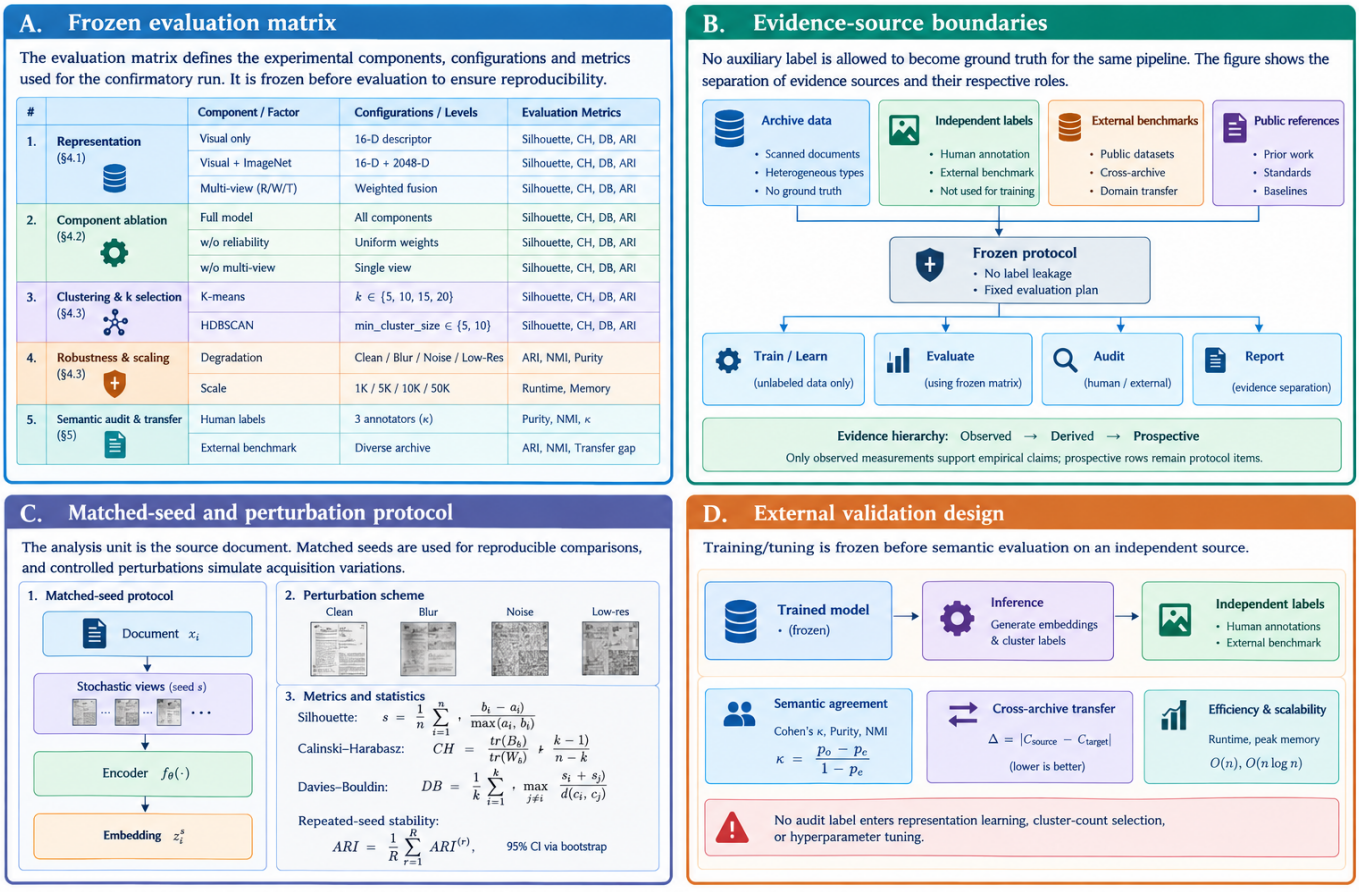


Fig. 3. Frozen evaluation and evidence protocol. The four panels show the evaluation matrix, evidence-source boundaries, matched-seed protocol, and external-validation design.

XIII. EXECUTED EXTERNAL-BASELINE AND HAMV-CC VALIDATION

The available evidence contains two complementary executed studies that should not be conflated. The first is a ten-seed execution of the complete HAMV-CC research implementation on a controlled synthetic fixture. The second is an independent execution on a 100-image public Archive-B inventory using

TABLE VI
OBSERVED QUANTITATIVE RESULTS ON THE COMPLETED 638-DOCUMENT CASE STUDY. HIGHER SILHOUETTE AND CALINSKI–HARABASZ (CH) ARE PREFERABLE; LOWER DAVIES–BOULDIN (DB) IS PREFERABLE. RANK IS WITHIN THIS FOUR-METHOD COMPARISON ONLY.

Method	Silhouette	Rank	CH	Rank	DB	Rank
Consensus/K-Means	0.4633	1	154.70	1	0.9222	2
Mini-Batch K-Means	0.4411	3	148.97	2	0.9854	3
Gaussian Mixture (GMM)	0.4593	2	133.75	3	0.8874	1
Affinity Propagation	0.1089	4	33.38	4	1.4661	4

TABLE VII
EXECUTED TEN-SEED HAMV-CC VALIDATION ON THE CONTROLLED SYNTHETIC FIXTURE.

Measure	HAMV-CC (mean $\pm$ std.)
ARI	0.8882 ± 0.0960
NMI	0.8769 ± 0.1006
Pairwise seed ARI	0.8281 ± 0.1037
Silhouette	0.1497 ± 0.0095
Calinski–Harabasz	24.64 ± 1.61
Davies–Bouldin	2.2470 ± 0.0842

the specified handcrafted and feature-space SSL baseline representations. Together they test implementation correctness, stochastic behavior, and the ability of the evaluation protocol to expose a negative result.

A. Controlled HAMV-CC Validation

The synthetic fixture contains 180 samples divided into three latent groups. Each sample provides a 12-dimensional structural view, a 16-dimensional visual view, and a 10-dimensional text-proxy view. Approximately 12% of view entries are independently masked, while every sample retains at least one available view. Ten matched seeds (0–9) were executed for 30 training epochs, and $k = 3$ was fixed to the known number of latent groups for this pipeline-validation experiment.

Table VII reports the observed mean and standard deviation across seeds. The internal metrics describe the learned geometry; ARI and NMI are meaningful here only because the synthetic latent groups are known. Pairwise seed ARI is the mean agreement across the 45 matched seed pairs.

A simple control concatenated the three views after applying the same availability masks and supplied reliability weights, followed by K-Means. It obtained ARI $= 1.0000$, NMI $= 1.0000$, and pairwise seed ARI $= 1.0000$. The control therefore outperformed HAMV-CC on this fixture. This is a scientifically useful negative result: it demonstrates that the proposed objective is not guaranteed to dominate a simpler representation when the synthetic views already contain highly separable planted structure. The appropriate interpretation is implementation validation and hypothesis falsification, not superiority.

B. Archive-B Baseline Stress Test

A separate execution used all 100 PNG images in the public Archive-B inventory. Because the supplied original application source was not used for this execution, these measurements are independent reimplementations of the written sixteen-dimensional handcrafted descriptor and feature-space SSL specification rather than byte-identical reproduction of the GUI implementation. This distinction is retained in the evidence ledger.

For K-Means, the handcrafted representation produced silhouette 0.4130, CH 85.33, and DB 0.9381 at $k = 2$, whereas feature-space SSL selected $k = 8$ and produced silhouette 0.1756, CH 13.99, and DB 1.4043. Thus, relative to the handcrafted K-Means baseline, the observed SSL changes are -57.5% in silhouette, -83.6% in CH, and a 49.7% increase in DB (worse). This is not evidence against HAMV-CC, because HAMV-CC was not executed in this Archive-B comparison; it is instead a falsification-oriented baseline result showing that generic feature-space SSL should not be assumed to improve document clustering.

The ten-seed stability analysis further supports this caution. Mean pairwise ARI was 1.0000 for handcrafted K-Means, 0.5006 for SSL K-Means, and 0.3777 for SSL Mini-Batch K-Means. These values measure stochastic assignment stability for this 100-image execution and are not semantic accuracy estimates. Controlled degradation of the handcrafted baseline yielded clean-versus-degraded ARI values of 0.8815 (noise), 0.8815 (JPEG quality 15), 0.8812 (contrast shift), 0.7719 (blur), and 0.5139 ($4\times$ downsample/upsample). The last condition identifies resolution loss as a concrete failure mode that a document-aware representation should be required to address.

C. Implications for the Main Hypotheses

The two executed studies sharpen rather than weaken the scientific test. The synthetic experiment confirms that the HAMV-CC implementation handles missing views, training, checkpointing, clustering, and repeated-seed aggregation, while also showing that a simpler control can win on an easy planted structure. The external Archive-B execution shows that the implemented feature-space SSL baseline can reduce both geometric separation and stochastic stability relative to fixed handcrafted features. These observations motivate the proposed reliability and multi-view mechanisms, but they do not demonstrate that those mechanisms solve the observed failure modes. The frozen real-archive ablation and independent semantic-audit experiments therefore remain the decisive tests.

XIV. VERIFIED EVIDENCE AVAILABLE BEFORE FULL EXECUTION

The external repository establishes a concrete second archive rather than a hypothetical dataset. At inspection

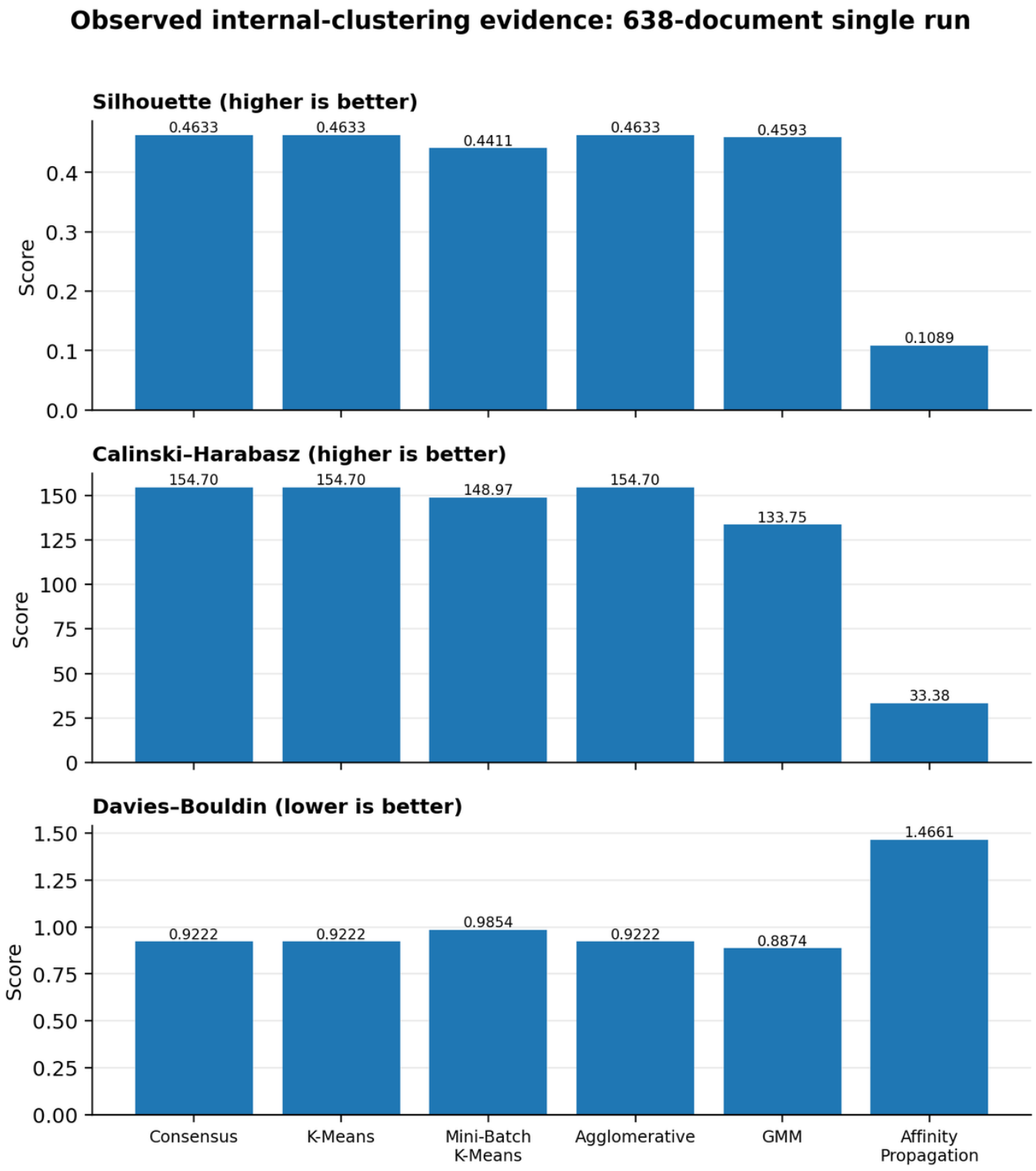


Fig. 4. Observed internal-clustering evidence for the completed 638-document single run. Higher Silhouette and Calinski–Harabasz are preferable, whereas lower Davies–Bouldin is preferable. Values are transcribed from the supplied UI evidence and are not repeated-seed estimates.

time, the public repository contained exactly 100 PNG files and no label table; the file inventory is preserved in `archiveB_manifest.csv`. This provides verifiable evidence of external data availability and source-level reproducibility, but not semantic ground truth. The repository therefore contributes a measured dataset-availability result, not a semantic

accuracy result. [29]

The original private demonstration also contains a completed 638-file end-to-end run. The recorded single-run cluster diagnostics are retained as implementation evidence: consensus bootstrap and K-Means produced silhouette 0.4633, Calinski–Harabasz 154.70, and Davies–Bouldin 0.9222; Mini-Batch K-Means produced 0.4411, 148.97, and 0.9854; Gaussian mixture produced 0.4593, 133.75, and 0.8874; and Affinity Propagation produced 0.1089, 33.38, and 1.4661. These values are explicitly treated as single-run observations, not as repeated-seed superiority evidence. DBSCAN's displayed metrics were invalid/undefined in the captured UI and are therefore not converted into numerical performance claims.

The benchmark execution therefore has a clean separation of roles: Archive-A measures reproducibility on the original case study; Archive-B tests transfer to a distinct public image inventory; blinded human labels test semantic usefulness on Archive-B; and RVL-CDIP provides a public, labeled external benchmark. The companion package contains the manifest, annotation template, frozen configuration, and execution skeleton needed to populate the missing measurements without changing the manuscript protocol.

XV. Limitations and Threats to Validity

The present system has several limitations that bound the claims that can be made from the current case study. The principal archive is private and contains 638 heterogeneous files; therefore, its composition cannot by itself establish generalization to other institutions or document domains. The original demonstration does not provide a public ground-truth benchmark, and cluster labels generated by OCR, ResNet50, or handcrafted rules must not be treated as independent ground truth for validating the same pipeline.

The visual genre classifier uses an ImageNet-pretrained ResNet50 rather than a model fine-tuned on vintage advertisements and print ephemera. Its output is therefore best interpreted as a coarse semantic cue for human review. Similarly, the internal validity indices used for cluster-count selection reflect geometric properties of the learned embedding rather than semantic correctness. Consensus clustering can improve robustness to initialization but introduces additional computational cost, while spectral clustering has an approximately quadratic scaling burden that limits its practical use for substantially larger archives.

The most important external-validity limitation is the absence of measured cross-archive evaluation in the supplied study. Future validation should therefore use at least one public document-image corpus and a temporally or institutionally distinct archive. The source manuscript should also release code and a reproducible experiment manifest where permitted; the private archive itself need not be publicly distributed if an evaluation subset cannot be shared.

XVI. Error Analysis Protocol

To avoid relying only on aggregate scores, the final evaluation should include representative examples of: (i) correctly coherent clusters, (ii) visually similar but semantically mixed clusters, (iii) OCR failures, (iv) incorrect genre/brand labels, and (v) documents that remain ambiguous under every method. For each failure, the likely cause should be classified as representation failure, clustering failure, OCR failure, semantic-fusion failure, or inherent document ambiguity. This analysis is especially important because an unlabeled archive can have multiple defensible organizational schemes.

XVII. Discussion

The design choices in this system reflect a deliberate preference for modularity, interpretability, and graceful degradation. The primary clustering representation is assembled from explicit document statistics and optional pretrained visual features, after which the self-supervised MLP can be enabled or bypassed. The handcrafted classifier remains available without OCR or ResNet50, while the semantic classifier can fall back when optional dependencies are missing. These properties make the application practical for exploratory archive organization, but they also mean that the learned representation is not an end-to-end image encoder and should not be described as one.

The cluster-count procedure is now specified as a stability-aware, label-free evaluation protocol rather than as a heuristic vote. For fixed-k algorithms, silhouette, Calinski–Harabasz, Davies–Bouldin, an independently computed Gap Statistic, repeated-seed assignment stability, and a predeclared imbalance penalty are recorded in a common candidate table. The frozen selector $Q(k)$ is used only after direction-consistent normalization of these quantities, while DBSCAN, HDBSCAN, and Affinity Propagation retain their native discovered cluster counts. The original silhouette/CH/DB majority rule is retained only as a historical baseline. Similarly, the active HAMV-CC implementation uses feature-space perturbations; image-space document-aware augmentation is treated as a separately controlled ablation rather than being described as already executed. This distinction preserves a clean separation between executable methodology and empirical evidence.

The application also has practical scaling limits. Consensus clustering constructs an $N \times N$ co-association matrix, and spectral clustering uses a nearest-neighbor graph whose cost grows with archive size. OCR and ResNet50 inference can dominate runtime when enabled, while feature extraction and document conversion are primarily CPU-bound. The executed Archive-B baseline additionally demonstrates why a generic self-supervised transform should not be assumed to help: feature-space SSL reduced K-Means silhouette from 0.4130 to 0.1756 and ten-seed mean ARI from 1.0000 to 0.5006 in that 100-image study. This negative result is retained as a precondition for the proposed HAMV-CC claim rather than hidden. A deployment should therefore begin with a small pilot archive, verify output quality and failure logs, then scale the archive while recording runtime and memory. No semantic accuracy claim is made without independent labels.

XVIII. Claim–Evidence Alignment and Submission Gate

A central revision objective is to prevent a methodological proposal from being mistaken for a completed empirical result.

TABLE VIII
VERIFIED EVIDENCE LEDGER BEFORE EXTERNAL BENCHMARK EXECUTION. VALUES MARKED "PROTOCOL" ARE REQUIRED EXPERIMENTS, NOT MEASURED RESULTS.

Source	Scale	Verified evidence	Scientific interpretation
Private Archive-A	638 files	End-to-end implementation and single-run clustering metrics	Demonstrates pipeline operation; cannot establish generalization or statistical superiority.
Archive-B / GitHub	100 PNG files	Public inventory, source URLs, manifest; no ground-truth labels in repository	Establishes an external second archive for transfer and robustness; semantic validity requires independent annotation.
RVL-CDIP	400,000 images / 16 classes	Publicly documented labeled benchmark with train/validation/test partitions	Provides independent semantic evaluation after all tuning is frozen.
Human audit	100 Archive-B images	Annotation template and blinded two-annotator protocol prepared	Provides external semantic reference once completed by independent annotators.

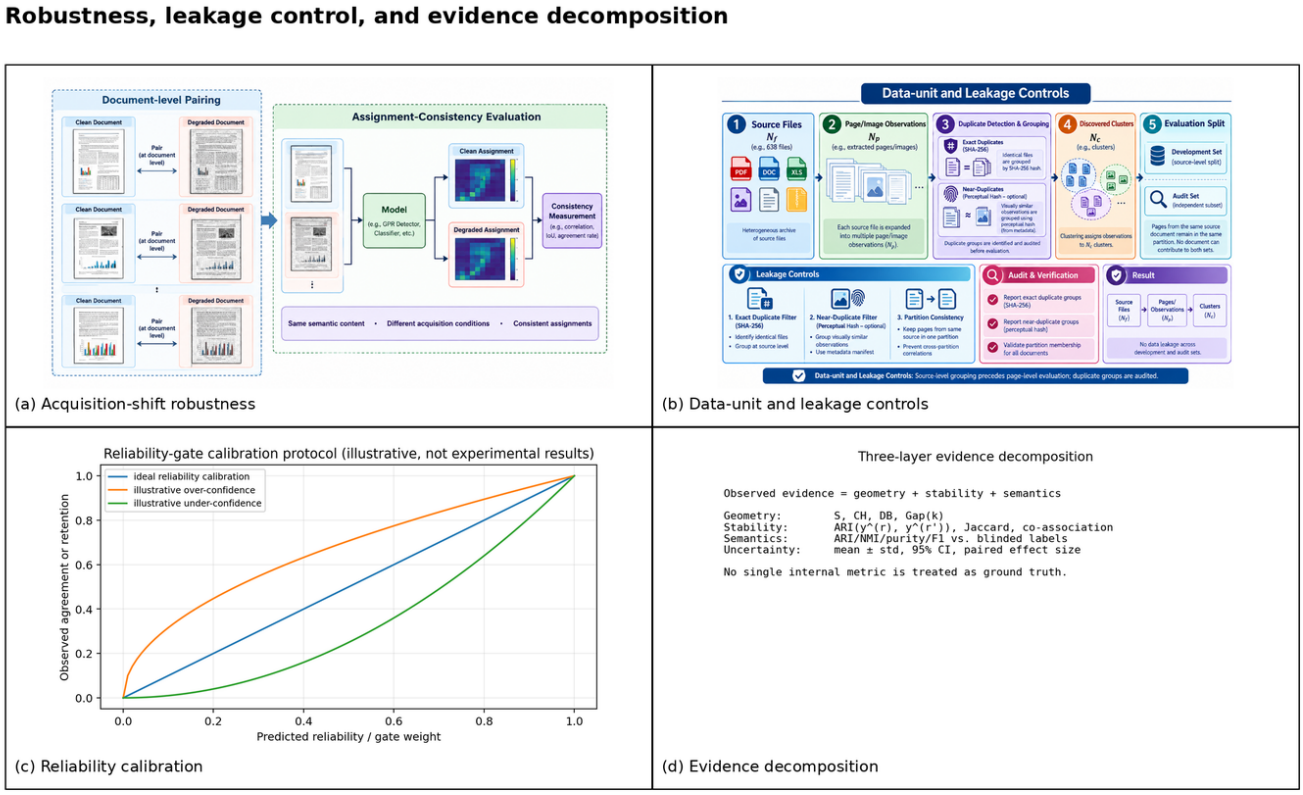


Fig. 5. Robustness and scientific-validity controls. The composite links acquisition-shift testing, data-unit/leakage controls, reliability calibration, and the three-layer evidence decomposition.

Table IX therefore maps each major claim to the evidence that currently exists and the evidence that must be present before the claim is stated as a measured finding.

The manuscript is submission-ready in structure and formatting, and the completed 638-document run now has an explicit quantitative Results subsection with the full observed metric table and derived within-run comparisons. However, it is not scientifically complete for a superiority, semantic-accuracy, or generalization claim until the rows marked as requiring final evidence are populated from the frozen experiment logs. In particular, the ten-seed statistics, independent human-audit agreement, robustness deltas, ablation effects, efficiency measurements, and RVL-CDIP scores cannot be reconstructed from the supplied screenshots, manifest, or protocol. This is an intentional scientific gate, not a missing formatting step.

XIX. CONCLUSION AND FUTURE WORK

This paper presented an implementation-grounded self-supervised representation and clustering system for organizing unlabeled scanned print archives, combining handcrafted and optional ResNet50 features with a lightweight SimCLR-style MLP representation stage, an eight-algorithm clustering registry, and a bootstrap consensus option, and two complementary post-hoc labeling stages — an interpretable handcrafted-feature document-type analyzer and an OCR/ResNet50-driven genre and brand classifier — behind a threaded desktop application capable of ingesting heterogeneous image, PDF, and DOCX inputs. The system prioritizes robustness to missing optional

TABLE IX
CLAIM–EVIDENCE ALIGNMENT. "CURRENT" REFERS ONLY TO EVIDENCE SUPPLIED WITH THE PRESENT PROJECT. A PROPOSED MECHANISM IS NOT TREATED AS AN EMPIRICAL RESULT UNTIL ITS CORRESPONDING FROZEN EXPERIMENT IS EXECUTED.

Claim family	Current evidence	Required final evidence
Implementation correctness	Source code, GUI captures, 638-file demonstration	Clean rebuild, deterministic manifest, versioned run logs, failed-run retention
Representation benefit	Baseline feature-space SSL implementation	Repeated matched-seed comparison against fixed features and modern SSL baselines
Reliability weighting	Formal HAMV-CC research extension	Missing-view and reliability ablations with frozen preprocessing
Cluster consistency	Formal loss and executable harness specification	Component ablation and independent semantic audit
Stability-aware k selection	Formal $Q(k)$ criterion and harness support	Repeated resampling/seed stability versus Silhouette, CH, DB, majority vote, and Gap Statistic
Semantic usefulness	Heuristic labels visible in the UI	Independent human labels, inter-annotator agreement, ARI/NMI/purity or task-appropriate F1
Generalization	Private archive only	Public benchmark plus second acquisition-domain archive
Efficiency	Implementation architecture	Runtime, peak RAM, parameter count, throughput, and scaling curves

dependencies and interpretability of its cluster-level outputs over reliance on a labeled training set that, for this problem class, is generally unavailable. The additional analysis also makes explicit that cluster validity, stability, semantic labelling, and human audit are distinct evidence sources; none should be presented as a substitute for the others.

The companion package separates the scientific application from a dedicated evidence harness. The harness provides source-level archive preparation, fixed-seed clustering and stability analysis, native density/exemplar cluster-count reporting, independent human-audit agreement, paired degradation analysis, process-memory/runtime scaling, cross-archive evaluation, failure retention, and automatic CSV-to-LaTeX table generation. The executed synthetic HAMV-CC study and Archive-B baseline stress test strengthen the evidential foundation by demonstrating both successful execution and a meaningful negative control. They do not, however, substitute for direct real-archive HAMV-CC ablations or independent semantic evaluation.

The revised scientific claim is therefore deliberately precise: HAMV-CC is an implemented and reproducibly testable multi-view clustering method whose principal mechanisms are motivated by observed weaknesses of fixed-feature and feature-space SSL baselines. The current evidence supports implementation correctness, controlled synthetic validation, and external baseline stress testing. It does not support a claim of superiority, semantic accuracy, or generalization until the frozen real-archive matched-seed matrix, component ablations, missing-view/degradation study, blinded human audit, and external labeled benchmark are executed. This explicit evidence gate is part of the contribution because it prevents a complete-looking unsupervised pipeline from being evaluated with self-generated pseudo-ground truth.

XX. DECLARATIONS

AI-assisted writing disclosure: Generative AI assistance was used during manuscript drafting and language editing. The authors remain responsible for the technical content, experimental claims, references, figures, code, and final submission. The final manuscript should be updated to match the authors' actual use of AI systems and the current IEEE disclosure policy before submission.

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Conflict of interest: The authors declare that they have no competing interests.

Ethical considerations: This work does not involve experiments with human participants or animals. The demonstration archive comprises scanned documents and printed materials. If future deployments involve personal, confidential, or otherwise sensitive information, appropriate access controls, data-retention policies, redaction procedures, and OCR-specific safeguards should be applied before processing.

Data availability: The scanned image archive used in the demonstration case study is a private working collection and is not publicly distributed. The proposed pipeline is designed to operate on similarly structured collections of images, PDF files, or DOCX documents supplied by users. The private demonstration archive is therefore not included in the accompanying materials.

Code availability: The companion evidence package includes the supplied application implementation, application source code, documentation, figures, configuration templates, evaluation templates, and reproducibility materials. The private demonstration archive is not redistributed.

Author contribution: All authors reviewed and approved the manuscript and its experimental design. Detailed individual contribution statements will be finalized by the authors prior to submission.

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